

Proton Stability from Occupant Uniqueness

An AASC Exhaustion Theorem for Baryon-Violating Transition Roles

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Abstract

This paper proves a downstream AASC constraint-theoretic proton-stability theorem, conditional on the locked AASC Standard-Model structural-recovery and occupant-uniqueness results. Proton decay is treated as a proposed standing-bearing transition role, not as a proton-lifetime parameter, numerical correction, or missing coefficient in an otherwise fixed table. The target domain is the unique AASC-admissible particle-physics bookkeeping occupant supplied by the locked Standard-Model occupant classification. A proton-decay carrier would have to destroy an admitted localized proton persistence class and introduce a transition ledger with nonzero baryonic-role loss. The proof proceeds through an explicit proton persistence certificate, a witnessed baryonic persistence-role ledger, quotient-morphism transition rigidity, and event-level target projection. A larger-domain BSM, GUT, EFT, quantum-gravity, or topological proposal is typed by target projection: if it induces no target transition witness, it does not license proton decay at the target level; if it does induce such a witness, it moves the target transition ledger and is not a same-target occupant. The theorem therefore establishes occupant-level proton stability: no proton-decay transition role is admissible in the unique target AASC particle-physics occupant. Conventional proton-decay theory and experimental-search literature is used only for orientation; the AASC proof does not compute a lifetime or replace phenomenological searches.

Keywords: AASC; proton decay; proton stability; baryon-violating operator; transition ledger; quotient morphism; event projection; Standard Model uniqueness; admissible re-description; standing-preserving construction; particle persistence class; role occupant; GUT; BSM; charge lattice; structural physics.

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1 Introduction: Proton Decay as a Transition-Role Question

1.1 The ordinary question and the AASC question

In conventional particle physics, proton decay is usually discussed as a dynamical or phenomenological question: does a theory contain baryon-violating interactions that produce a finite proton lifetime, and what lower bound has experiment placed on the corresponding partial lifetime? In grand-unified or other enlarged frameworks, a proton-decay channel is typically associated with a heavy mediator or an effective higher-dimensional operator. That question is legitimate, but it is not the question answered here.

AASC asks a prior structural question. Suppose one has a target particle-physics domain whose statements are reusable, transportable, boundary-detectable, and standing-preserving. Suppose further that the AASC classification theorem has identified a unique admissible occupant for that domain. Can a proton-decay channel be added as a same-target redescription, a parameter value, an EFT skin, or a hidden branch? Or would it be a new structural transition role?

The answer defended in this manuscript is that proton decay is not a parameter. It is not analogous to choosing a value of a gauge coupling or a Yukawa eigenvalue. A decay channel would require a transition carrier that destroys an admitted proton persistence class:

$$[p]_{\text{persist}} \longrightarrow [F]_{\text{persist}}, \quad [p]_{\text{persist}} \not\subset [F]_{\text{persist}}.$$

If that transition is physically standing, it must appear in the target transition ledger. If it appears, the target occupant has changed. If it does not appear, the claimed channel is not an admissible target transition. If it is asserted without a witness, it is a hidden selector or repair.

1.2 Why the issue is structural rather than numerical

The recovered standalone Standard Model structure in the supplied AASC material is a structural tuple. It includes gauge-product, representation-response, Higgs/bridge, Yukawa-realizability, charge-lattice, anomaly, vacuum, mass-response, mixing/phase, and family-cardinality entries. It explicitly separates those structural entries from numerical parameter values. Changing an operator-availability ledger or transition ledger changes the structural object, not merely a value inside it.

A proton-decay carrier is therefore not a small correction to an already present channel. There is no target proton-decay rate waiting to be assigned a tiny number. The channel itself must be admitted before a rate can be meaningful. The relevant question is:

$$O_{\Delta B} \in \mathcal{O}_{\text{target}} ?$$

where $O_{\Delta B}$ denotes a baryon-role-changing transition-operator class. If $O_{\Delta B}$ is absent from the target occupant, then assigning a coefficient to it is not parameter completion. It is load motion or witness motion.

1.3 The bridge beyond same-interior closure

A same-interior Standard Model argument would say only this:

$$\Sigma_{\text{SM}} \Rightarrow \text{no internal renormalizable proton-decay operator.}$$

That is too weak for the present goal. The supplied AASC capstone material distinguishes fixed-load interior closure from occupant-level uniqueness. Fixed-load closure says that, inside an already declared Standard-Model bookkeeping interior, no further same-domain completion remains. Occupant uniqueness says that the admissible target particle-physics bookkeeping occupant is unique up to admissible equivalence.

The present paper uses both levels. Structura-style fixed-load closure blocks same-interior addition of a baryon-violating operator as a redescription, parameter, or hidden repair. Occupant-level uniqueness blocks a non-equivalent target occupant containing a standing proton-decay transition role. The new work of this paper is the bridge: a physically standing proton-decay claim, even if generated by a larger-domain parent, must project to a target transition witness if it is genuinely a target proton decay.

1.4 Proof strategy

The proof has seven steps.

1. Certify the proton as an admitted target persistence role, while refusing to derive hadron spectroscopy or proton mass in this paper.
2. Define the baryonic role ledger by witnesses rather than by a primitive global conservation postulate.
3. Show that $\Delta B_{\text{role}} \neq 0$ is transition-ledger motion.
4. Prove that same-target admissible transitions must be quotient morphisms licensed by the target transition ledger.
5. Show that proton decay is not such a licensed target quotient morphism.
6. Show that larger-domain decay claims either project to target transition witnesses or do not license target proton decay.
7. Apply occupant uniqueness to exclude any non-equivalent target occupant containing a proton-decay role.

1.5 How to read the result

The strongest compact reading is:

A proton-decay channel is a new transition role;
the target occupant is already exhausted.

The result is stronger than the ordinary statement that the renormalizable Standard Model does not contain an observed proton-decay interaction. It is weaker than the unconstrained metaphysical statement that no imaginable mathematical theory could encode proton decay. That is the correct AASC posture: strong within the target standing domain, typed rather than dismissive toward larger-domain model building.

2 Claim boundary and falsification contract

The paper’s title-level claim must be read with its AASC type attached. The theorem says that no proton-decay transition role is admissible in the unique target AASC particle-physics occupant. It does not say that no formal syntax can write a baryon-violating term, and it does not say that no larger-domain model can be proposed.

Definition 2.1 (Target proton-decay claim). A target proton-decay claim is a claim that there exists an admissible standing-bearing transition carrier

$$T_{p \rightarrow F}^{\text{Adm}} : [p]_{\text{persist}} \longrightarrow F$$

such that F lacks the original proton persistence class and

$$\Delta B_{\text{role}}(T_{p \rightarrow F}^{\text{Adm}}) \neq 0.$$

The word “target” means that the transition is claimed inside the particle-physics standing quotient, witness family, transport structure, continuation structure, and role ledger of the AASC target occupant.

Nonclaim 2.2 (What is not being asserted). This paper does not claim any of the following: no formal larger-domain Lagrangian can contain a $\Delta B \neq 0$ term; no BSM or GUT proposal can be formulated; no quantum-gravity or nonlocal proposal can be discussed; no experiment can report an anomalous event; AASC computes a proton lifetime; baryon number is an unanalysed primitive conservation law. The claim is structural and target-domain: no proton-decay transition role is admitted in the unique target occupant.

Criterion 2.3 (Falsification contract). *A reproducible empirical proton-decay claim would not be ignored by the theorem. It would force at least one of the following outcomes:*

1. *the event is not a boundary-detectable loss of the target proton persistence class;*

2. the event induces a new target transition witness, hence larger-domain target-ledger motion;
3. one of the locked AASC dependencies used by the occupant-uniqueness theorem fails;
4. the quotient-morphism transition-rigidity theorem fails.

Thus the theorem is not insulated from evidence. It fixes what a proton-decay observation would have to change.

Remark 2.4 (Same quotient versus induced witness). The central discriminator for larger-domain proposals is:

$$L^+|_{\text{tar}} = \begin{cases} \text{SM}_{\text{book}}, & \text{same quotient; no target proton-decay role,} \\ \text{SM}_{\text{book}} + W_T^{\Delta B}, & \text{induced transition witness; target ledger motion,} \\ \perp, & \text{standing failure or no target projection.} \end{cases}$$

This formula appears repeatedly because it is the shortest safe expression of the BSM/GUT/UV relation. A larger-domain proposal is physically interesting only if it has a target projection. Once projected, either it leaves the target quotient unchanged or it induces a new target witness.

Protocol 2.5 (Use of the word “stability”). *When the manuscript says “proton stability”, it means occupant-level target stability: no admissible transition role in the unique target occupant destroys the admitted proton persistence class. It does not mean a proof that every imaginable external formalism lacks a symbol resembling proton decay. This protocol is binding throughout the paper.*

3 Terms, notation, and proof roadmap

3.1 Core terms

Definition 3.1 (Standing). Standing is identity-bearing status preserved under admissible redescription, transport, continuation, and reuse. A distinction that has no invariant witness inside the declared target domain is skin or hidden load; it is not standing-bearing target content.

Definition 3.2 (Bookkeeping occupant). A bookkeeping occupant is not a Lagrangian alone and not a coordinate chart. In the AASC classification it is a tuple of parameter/label space, scopes, outputs, witness family, standing equivalence relation, licensed transports, histories, role typing, and boundary/envelope ledger. In schematic form:

$$B = (P, S, Out, W, \sim, T, H, \rho, E).$$

Occupant equivalence means identity-preserving equivalence on the standing quotient, witnesses, transports, histories, role typing, and boundary ledger.

Definition 3.3 (Target persistence quotient). Let $\mathcal{P}_{\text{tar}}$ denote target particle-persistence records. The persistence standing quotient is

$$\mathcal{P}_{\text{tar}} / \sim_{\text{pers}} .$$

A particle-persistence class is written $[X]_{\text{persist}}$. The proton persistence class is $[p]_{\text{persist}}$.

Definition 3.4 (Transition ledger). The target transition ledger is

$$\mathcal{T}_{\text{tar}} = (\mathcal{P}_{\text{tar}}, \mathcal{O}_{\text{tar}}, W_T, \sim_T, H_T, \rho_T, E_T),$$

where $\mathcal{O}_{\text{tar}}$ is the admitted transition/operator carrier family, W_T the transition witnesses, $\sim_T$ the transition equivalence relation, H_T the admissible history/continuation structure, ρ_T the anchor/tensor/skin role typing, and E_T the boundary/envelope status.

Definition 3.5 (Baryonic role). The baryonic persistence role B_{role} is not primitive baryon number. It is a witnessed composite persistence-role coordinate defined by a target witness family W_B . A proton-decay transition is a transition for which $\Delta B_{\text{role}} \neq 0$ and the final family lacks the original proton persistence class.

3.2 Proof roadmap

The proof proceeds in seven steps:

1. certify the proton persistence role as an admitted target persistence class;
2. define B_{role} by witnesses rather than by a primitive conservation postulate;
3. prove that $\Delta B_{\text{role}} \neq 0$ is transition-ledger motion;
4. prove that same-target transitions must descend to ledger-licensed morphisms on standing quotients;
5. show that a proton-decay morphism is not licensed by the target transition ledger;
6. prove that larger-domain decay claims must project to target witnesses or fail to count as target decay;
7. apply occupant uniqueness to block a non-equivalent target occupant.

3.3 Notation

Symbol	Meaning
SM_{book}	unique target Standard-Model bookkeeping occupant
B_{AASC}	AASC admissible particle-physics bookkeeping space
$[p]_{\text{persist}}$	admitted proton persistence class
B_{role}	witnessed baryonic persistence-role coordinate
W_B	witness family defining B_{role}
$\mathcal{T}_{\text{tar}}$	target transition ledger
$T_{\#}$	quotient-level transition morphism
$\pi_{\text{tar}}(L^+, e)$	target projection of a larger-domain event
$O_{\Delta B}$	baryon-violating operator/transition class
$X_{\Delta B}$	heavy mediator or GUT-type parent carrier

4 Dependency ledger and imported authority

The proof is downstream of the supplied AASC corpus. The present paper does not rederive the Standard Model structural tuple, the fixed-load interior closure, charge-lattice rigidity, or occupant uniqueness. It imports them with specific roles.

Imported source	Imported authority	Use in this paper
Standalone Standard Model Structure	The recovered SM object is a structural tuple, with gauge product, representation skeleton, Higgs/bridge carrier, Yukawa/operator availability ledger, charge lattice, anomaly ledger, vacuum form, mass/mixing slots, and family record; numerical values remain downstream.	Establishes that adding a proton-decay operator is structural operator-ledger motion rather than parameter motion.
Structura Ex Necessitate Standardis Mod- elis	Fixed-load interior closure; added-load dichotomy; same-domain counterexample burden; no fifth same-scope locus; higher-dimensional operators only as transport images rather than independent primitives.	Blocks same-load completion and classifies primitive ΔB operators as new load unless they are certified transport images.

Classification Theorem	Defines bookkeeping occupants; proves singleton occupant classification up to admissible equivalence; supplies terminal/standing-conservation style closure in the target particle-physics bookkeeping space.	Supplies final occupant-level exclusion once transition-ledger motion is proved.
SR7 realization rigidity	Product, representation, Abelian charge, anomaly, residual/vacuum survivor ledgers assemble into a unique role-equivalence class inside the gauge-realization comparison regime.	Blocks hidden gauge-skeleton substitution as a route to target proton decay.
Charge-lattice rigidity and CS6A	Charge assignments are unique up to normalization/sign in fixed charge scope; continuous deformation, additive offset, hidden selector, extra $U(1)$, extra chiral matter, and extra Higgs are same quotient, inadmissible, or larger scope.	Supplies the route-exhaustion pattern for transition ledgers and prevents charge-normalization routes from becoming decay carriers.
Metric-relative reconstruction registry	Separates target admission, source constraint, shape closure, anchor status, and numerical reconstruction.	Reinforces that proton decay is not a numerical reconstruction target or lifetime slot before transition admission.

Theorem 4.1 (Dependency separation). *No single imported dependency is identical to the proton-stability theorem. The standalone structure paper supplies the structural/parameter boundary; Structura supplies fixed-load and same-domain closure; the Classification Theorem supplies occupant uniqueness; the charge papers supply a route-exhaustion pattern; this manuscript supplies the proton-specific transition-ledger and event-projection bridge.*

Proof. If the standalone structural tuple alone were sufficient, the paper would prove only that $O_{\Delta B}$ is not part of the displayed tuple. If Structura alone were sufficient, the paper would prove only fixed-load interior closure. If the Classification Theorem alone were sufficient, the paper would not explain why proton decay is a rival occupant rather than a larger-domain event. The new bridge proves that physically claimed proton decay either induces target transition-ledger motion or does not license target decay. Only after that bridge is occupant uniqueness decisive. $\square$

5 The proton persistence certificate

The theorem does not use the word “proton” as an unanalyzed primitive. The proton role used here is a localized composite persistence role admitted inside the target occupant under an explicit certificate. The paper does not derive confinement, the proton mass, the hadron spectrum, nuclear binding, or nonperturbative QCD matrix elements. Its question is narrower and precisely typed: once the target occupant contains a boundary-detectable proton persistence class, can an admissible target transition destroy that class?

5.1 Role package and admission boundary

Definition 5.1 (Proton role package). The proton role package is

$$P_p = (\Pi_{\text{loc}}, C_{\text{neutral}}, Q_{+1}, M_{\text{resp}}, \partial_{\text{detect}}, B_{\text{role}}),$$

where Π_{loc} is localized persistence standing, C_{neutral} is color-neutral composite/hadronic role status, Q_{+1} is the charge-lattice presentation of electric charge +1, M_{resp} is mass/inertial response standing, ∂_{detect} is the boundary-detectable identity record, and B_{role} is the baryonic persistence-role coordinate witnessed in Section 6.

Definition 5.2 (Admitted proton persistence class). The admitted proton persistence class is the standing class

$$[p]_{\text{persist}} \in \mathcal{P}_{\text{tar}} / \sim_{\text{pers}}$$

whose target role package is P_p . Two presentations of the proton role lie in the same class exactly when they preserve the localized composite identity, charge-lattice display, mass-response role, boundary identity, and baryonic-role witnesses under admissible redescription.

The role package is deliberately composite. It is not a claim that the proton is fundamental in the same sense as the chiral field roles of the structural tuple. It is a downstream persistence class over the recovered target occupant. The distinction matters: the present theorem proves stability of an admitted target persistence class, not first-principles derivation of the entire QCD bound-state spectrum.

5.2 Hadronic persistence admission

Definition 5.3 (Hadronic persistence admission datum). A hadronic persistence admission datum for a target class $[H]_{\text{persist}}$ is

$$\text{HadAdm}([H]_{\text{persist}}) = (D_{\text{phys}}, G_{\text{SM}}, Q_{\text{lat}}, M_{\text{slot}}, \partial_H, H_{\text{hist}}, \rho_H),$$

where D_{phys} is the realized physical interior, G_{SM} is the recovered product/representation/charge environment, Q_{lat} is the charge-lattice display carried by the class, M_{slot} is its mass-response role, ∂_H is its boundary identity record, H_{hist} is the

admissible history family under which the class is reused, and ρ_H records its role typing as downstream composite persistence rather than new fundamental load.

Lemma 5.4 (Hadronic Persistence Admission). *Within the recovered target Standard-Model structural occupant, a localized color-neutral composite class with a hadronic persistence admission datum is admitted as a downstream target particle-persistence class. The proton role $[p]_{\text{persist}}$ is admitted for this theorem through the datum $\text{HadAdm}([p]_{\text{persist}})$, not by an informal appeal to ordinary language.*

Proof. The admission rests on three imported structures and one local certificate. First, the realized-physics layer admits fields as persistence carriers and particles as localized persistence classes. Second, the recovered Standard-Model structural occupant supplies the gauge/product ledgers, representation-response skeleton, charge lattice, anomaly/vacuum compatibility, mass-response slots, and boundary-comparison discipline. Third, the charge-lattice and mass-response ledgers allow a downstream composite target to be displayed with electric charge, inertial response, and boundary-detectable identity without promoting that composite to a new fundamental field role.

The local certificate is $\text{HadAdm}([p]_{\text{persist}})$. It records that the class is color-neutral at the target interface, has the charge-lattice presentation Q_{+1} , has a mass-response role, and remains reusable under ordinary admissible target histories such as scattering, dressing, detector transport, and environmental embedding. These histories need not compute a binding energy. They only certify that the same target object can be reidentified across admissible presentations.

Thus the proton class is not inserted as a hidden primitive. It is a downstream localized composite persistence class over the already admitted target occupant. Loss of the admission datum is not a different skin of the same proton; it is a transition claim requiring a ledger-licensed quotient morphism. $\square$

Remark 5.5 (Conditionality and dependency typing). The lemma is conditional in the precise sense required for an auditable theorem. If a critic denies that the target domain contains any boundary-detectable proton persistence class, then the proton-decay question has been removed from the target problem. If the class is admitted, the present theorem audits whether a transition can destroy it. The proof does not require first-principles derivation of the proton mass, confinement mechanism, hadronic matrix elements, or full hadron spectrum.

5.3 Persistence, interaction, and decay

Definition 5.6 (Persistence-preserving process). A target process U is proton-persistence-preserving when

$$U_{\#}([p]_{\text{persist}}) = [p]_{\text{persist}}$$

up to admissible redescription of boundary skins, detector records, dressing, or scattering products. Such a process may alter kinematic display, environment, or emitted radiation, but

it does not destroy the proton role package P_p .

Definition 5.7 (Proton-decay process). A target process is a proton-decay process only if its quotient action sends the admitted proton persistence class to a final family lacking the proton role package:

$$T_{\#}([p]_{\text{persist}}) = [F]_{\text{persist}}, \quad [F]_{\text{persist}} \not\prec_{\text{Adm}} [p]_{\text{persist}}, \quad P_p \not\subseteq P_F.$$

In the notation of the later sections this is the target case $\Delta B_{\text{role}} \neq 0$.

Theorem 5.8 (Proton Persistence Certificate). *Given the realized physical interior, the recovered target Standard-Model structural occupant, and the hadronic persistence admission datum $\text{HadAdm}([p]_{\text{persist}})$, the proton role P_p defines a standing-bearing target persistence class $[p]_{\text{persist}}$. Any process preserving P_p is not proton decay. Any process destroying P_p must be licensed by a transition carrier whose standing is not supplied by proton persistence itself.*

Proof. The components of P_p are not optional labels. Localized persistence fixes the object as a target particle-persistence class; the color-neutral composite role distinguishes it from raw colored constituents; the charge-lattice presentation fixes its electric charge display; the mass-response slot fixes its inertial response status; boundary detectability fixes reusable empirical identity; and B_{role} fixes the baryonic composite-persistence coordinate by the witness family of Section 6.

If a process preserves these components up to admissible redescription, the proton class persists. The process may be ordinary scattering, dressing, recoil, bound-state embedding, or detector transport, but it is not proton decay because the standing class remains the same. If a process destroys one or more standing-bearing components of P_p , the process cannot be justified by the mere existence of P_p . It requires an independent transition carrier with a quotient action on target persistence classes. That carrier is exactly what the transition-ledger theorems audit. $\square$

Nonclaim 5.9 (No QCD numerical derivation). The certificate does not derive the proton mass, the confinement scale, the baryon spectrum, nuclear stability, lifetime bounds, or hadronic matrix elements. Those are downstream or sectoral questions. The certificate only fixes the object whose proposed decay is under audit.

6 Baryonic role ledger: witnessed role content, not a primitive conservation law

A proton-decay theorem must not smuggle in baryon-number conservation. The paper therefore does not treat B as a primitive metaphysical charge and does not use the conventional statement “baryon number is conserved” as a premise. Instead it defines a baryonic persistence role as a witnessed target bookkeeping coordinate. The role is standing-bearing only through

the witnesses that classify a baryonic composite persistence class inside the target occupant and distinguish that class from mesonic, leptonic, radiation, or vacuum-supported final families.

6.1 Witness family

Definition 6.1 (Baryonic persistence witness family). For a target persistence class $[X]_{\text{persist}}$, define

$$W_B([X]_{\text{persist}}) = (W_{\text{comp}}, W_{\text{color0}}, W_{\partial}, W_{\text{scatter}}, W_{\text{nonmes}}, W_{\text{nonlep}}),$$

where:

1. W_{comp} witnesses composite hadronic persistence role;
2. W_{color0} witnesses color-neutral target hadronic status;
3. W_{∂} witnesses boundary-detectable identity under the target interface;
4. W_{scatter} witnesses persistence under admissible scattering, dressing, and environmental histories;
5. W_{nonmes} distinguishes the role from mesonic composite final families;
6. W_{nonlep} distinguishes the role from leptonic final families.

The package W_B is a classification witness ledger. It is not a primitive global symmetry postulate and it is not a conservation rule.

Definition 6.2 (Baryonic role coordinate). The baryonic role coordinate is defined by witnesses:

$$B_{\text{role}}([X]_{\text{persist}}) = 1$$

iff the class $[X]_{\text{persist}}$ satisfies the baryonic persistence witness package W_B in the target occupant. It is undefined as standing-bearing content without that witness package. A purely conventional baryon label with no witness content is skin.

Remark 6.3 (Classification-first, not conservation-first). The witness package W_B is introduced to classify an already admitted composite persistence role, not to assert that the role cannot change. It distinguishes baryonic composite persistence from mesonic, leptonic, radiation, and vacuum-supported final families under the target interface. The no-decay result is not contained in this definition. It enters later, when a proposed loss of the witnessed role is tested against quotient-morphism transition rigidity and occupant uniqueness.

Lemma 6.4 (Independence of the baryonic witness package). *W_B is independent of the proton-stability conclusion. It is the minimal target witness package needed to identify a baryonic composite persistence class before any decay claim is considered.*

Proof. The components of W_B serve classification functions: composite hadronic role, color-neutral target status, boundary identity, persistence under admissible scattering and dressing, and distinction from mesonic or leptonic final families. None of these components says that the role cannot be lost. They only determine what it means, in the target occupant, for the baryonic role to be present. A decay claim is the separate assertion that a transition maps a class satisfying W_B to a family failing it. Therefore the witness package is classification-first and does not assume the theorem’s conclusion. $\square$

Definition 6.5 (Baryonic-role loss). A transition T has baryonic-role loss when its quotient action maps a class satisfying W_B to a final family that fails the package:

$$\Delta B_{\text{role}}(T) \neq 0 \iff W_B([p]_{\text{persist}}) \text{ holds and } W_B(T_{\#}[p]_{\text{persist}}) \text{ fails.}$$

This is a role-ledger condition, not a primitive conservation-law assertion.

6.2 Independence of the witness package

Remark 6.6 (Classification-first, not conservation-first). The package W_B is not defined from the conclusion that protons do not decay. It is the minimum witness package needed to classify a baryonic composite persistence role at the target interface: composite hadronic status, color-neutral target presentation, boundary identity, persistence under admissible histories, and distinction from mesonic or leptonic final families. The package classifies the role. It does not assert that the role cannot change. The impossibility of an admissible role-destroying transition is supplied later by quotient-morphism transition rigidity and occupant uniqueness.

Proposition 6.7 (Witness independence from the no-decay conclusion). *The baryonic witness package W_B can be stated before any stability theorem is invoked. It records the target conditions under which a class counts as baryonic at all; it does not contain a no-decay axiom.*

Proof. Each component of W_B is classificatory. W_{comp} and W_{color0} type the class as a color-neutral hadronic composite. W_{∂} and W_{scatter} type the class as a reusable boundary-detectable persistence class under admissible histories. W_{nonmes} and W_{nonlep} separate the baryonic role from other target final-family types. None says the role is indestructible. A transition that removes the role is logically definable; the later theorem asks whether such a transition is admissibly licensed inside the unique target occupant. $\square$

6.3 No hidden selector

Lemma 6.8 (Baryonic role is not a hidden selector). *B_{role} has standing only through W_B . A bare label “baryonic” with no witness content is skin; a label that changes a target witness is tensor motion; a label retained as authoritative without witnesses is hidden selector failure.*

Proof. AASC role typing exhausts anchor, tensor, and skin. If the label changes no witness in W_B , it does not change target standing and is skin. If it changes at least one witness in W_B , it is tensor content because an admitted target distinction changes. If it is asserted as identity-bearing while changing no witness, it is hidden unwitnessed structure. Thus B_{role} is not a fourth role and not a primitive conservation postulate. $\square$

Theorem 6.9 (Baryonic Role Standing). *The baryonic persistence role is standing-bearing only insofar as it is fixed by W_B . Loss of B_{role} is loss of witnessed persistence-role content, not loss of an arbitrary imposed label.*

Proof. By definition, B_{role} is fixed exactly by the witness package W_B . If all components of W_B are preserved under a proposed transition, then the baryonic persistence role persists. If B_{role} changes, at least one component of W_B changes: the composite hadronic role is lost, the boundary identity changes, persistence under admissible histories fails, or the final family is witnessed as mesonic or leptonic rather than baryonic. A witness change is not skin. Therefore $\Delta B_{\text{role}} \neq 0$ is target witness motion. $\square$

6.4 Role loss and transition ledgers

Proposition 6.10 (Witnessed role loss is not parameter motion). *A change in B_{role} cannot be represented as a change of a gauge coupling value, Yukawa value, mass parameter, basis coordinate, phase convention, charge normalization, or detector skin.*

Proof. Changing a numerical value inside an admitted occupant changes a coordinate or boundary value while preserving the occupant's witness family and transition ledger. Changing B_{role} changes whether the final standing family satisfies W_B . That alters the target persistence-continuation content. A coordinate choice can display the before and after records, but cannot create the transition witness joining them. Thus the change is structural transition-ledger motion. $\square$

Corollary 6.11 (Baryonic-role loss is transition-ledger motion). *If a transition T satisfies $\Delta B_{\text{role}}(T) \neq 0$, then the target persistence-continuation ledger changes:*

$$\Delta B_{\text{role}}(T) \neq 0 \quad \Rightarrow \quad \Delta \mathcal{T}_{\text{tar}} \neq 0,$$

unless the transition is already admitted as a ledger-licensed quotient morphism.

Proof. A change in B_{role} is a change in the witness family W_B for the affected persistence class. The transition ledger is precisely the structure that records admitted transitions among persistence classes, witnesses, histories, and role typing. Therefore role loss is transition-ledger motion unless it is already represented by an admitted transition morphism in that ledger. $\square$

7 Formalization of a proton-decay carrier

Definition 7.1 (Candidate proton-decay carrier). A candidate proton-decay carrier is a tuple

$$T_{p \rightarrow F}^{\text{cand}} = (D_p, F, \mathcal{O}_T, W_T, \tau_T, \partial_T),$$

where D_p contains the proton persistence class $[p]_{\text{persist}}$, F is the proposed final-state persistence family, $\mathcal{O}_T$ is the proposed operator or transition carrier, W_T is the transition-witness package, τ_T is redescription/transport data, and ∂_T is the boundary-detection record.

Definition 7.2 (Proton-decay status). A candidate $T_{p \rightarrow F}^{\text{cand}}$ is proton-decay-positive only if all of the following hold:

1. the domain includes $[p]_{\text{persist}}$;
2. the target projection of the final family lacks P_p ;
3. $\Delta B_{\text{role}}(T_{p \rightarrow F}^{\text{cand}}) \neq 0$;
4. the transition is claimed as standing-bearing rather than merely formal notation;
5. the transition has a transport-complete standing certificate.

If any item fails, the object is not an admissible target proton-decay carrier.

Lemma 7.3 (Decay channel versus decay rate). *A proton lifetime or decay rate is meaningful only after a decay transition role is admitted. A coefficient, scale, or lifetime parameter cannot create the channel whose rate it is supposed to measure.*

Proof. Let Γ_p denote a decay rate. If no transition carrier $T_{p \rightarrow F}$ is admitted, then Γ_p is not a value inside a target transition ledger; it is an attempted selector for a missing ledger entry. If $T_{p \rightarrow F}$ is admitted, then Γ_p may be a parameter of that admitted channel. Therefore channel admission is prior to lifetime assignment. $\square$

Proposition 7.4 (Structural typing of $O_{\Delta B}$). *A baryon-violating proton-decay operator $O_{\Delta B}$ is structural transition-role content. It is not gauge skin, basis skin, charge normalization, a Yukawa value, a mass correction, a finite counterterm, or an empirical anchor.*

Proof. Changing $O_{\Delta B}$ changes whether a transition class exists from $[p]_{\text{persist}}$ to a non-proton final persistence family. That changes the operator-availability ledger and the persistence-continuation ledger. By the structure/parameter criterion imported from the Standalone SM structural spine, changing an operator-availability ledger or transition ledger is structural. A numerical coefficient attached to an already-admitted $O_{\Delta B}$ would be parameter-like; admission of $O_{\Delta B}$ itself is not. $\square$

8 Transition-ledger rigidity

The central bridge theorem is not merely a case split. A same-target transition must act as a well-defined morphism on standing classes. A claimed decay that destroys the proton persistence class without a ledger entry is not a same-target morphism; adding the ledger entry changes the occupant. This section gives the quotient-morphism proof.

8.1 Target transition quotient

Definition 8.1 (Target transition quotient). The target transition quotient is the quotient of target persistence records by persistence standing equivalence:

$$\mathcal{P}_{\text{tar}} / \sim_{\text{pers}} .$$

An admitted transition must descend to a function on this quotient. The induced quotient action is denoted

$$T_{\#} : \mathcal{P}_{\text{tar}} / \sim_{\text{pers}} \longrightarrow \mathcal{P}_{\text{tar}} / \sim_{\text{pers}} .$$

Definition 8.2 (Target transition ledger). The target transition ledger is

$$\mathcal{T}_{\text{tar}} = (\mathcal{P}_{\text{tar}} / \sim_{\text{pers}}, \mathcal{O}_{\text{tar}}, W_T, H_T, \rho_T, E_T),$$

where $\mathcal{O}_{\text{tar}}$ is the admitted transition/operator carrier family, W_T is the transition witness ledger, H_T is the admissible continuation/history structure, ρ_T is anchor/tensor/skin role typing, and E_T is boundary/envelope status for transition claims.

Definition 8.3 (Ledger-licensed quotient morphism). A map $T_{\#}$ is ledger-licensed when it is represented in $\mathcal{T}_{\text{tar}}$, preserves the relevant standing quotient, carries the required transition witnesses in W_T , respects admissible histories H_T , and preserves role typing ρ_T except where an explicitly admitted transition changes a role.

8.2 Quotient descent

Lemma 8.4 (Transition descent criterion). *Let T be a target transition rule on representatives of persistence records. If T is admissible as a same-target transition, then*

$$x \sim_{\text{pers}} x' \implies T(x) \sim_{\text{pers}} T(x').$$

Consequently T descends to a unique quotient action $T_{\#}$ on $\mathcal{P}_{\text{tar}} / \sim_{\text{pers}}$.

Proof. Assume $x \sim_{\text{pers}} x'$ but $T(x) \not\sim_{\text{pers}} T(x')$. Then the result of applying the alleged transition depends on the representative chosen for the same standing class. A committed persistence record could be transported through one representative and yield a different final standing class than through another. That violates stable identity fixation and reuse of

construction records. Same-target admissibility therefore forces representative independence. The quotient action is then defined by $T_{\#}([x]) = [T(x)]$, and uniqueness follows from surjectivity of the quotient map. $\square$

Lemma 8.5 (Ledger licensing is stronger than quotient descent). *A quotient action $T_{\#}$ is not automatically an admissible transition. It is admissible in the target occupant only when represented in $\mathcal{T}_{\text{tar}}$ with the required witness, history, role, and boundary ledgers.*

Proof. A set-theoretic function on standing classes can always be written. AASC admissibility requires more: the action must preserve or lawfully transform the target witnesses, be transport-complete across admitted skins, be stable under histories, and not be generated by post-hoc representative selection. Those data are exactly what $\mathcal{T}_{\text{tar}}$ records. Therefore quotient descent is necessary but not sufficient; ledger licensing is the additional standing requirement. $\square$

8.3 Rigidity theorem

Theorem 8.6 (Quotient-Morphism Transition Rigidity). *Let T be a claimed same-target transition on target particle-persistence classes. If T is admissible as a same-target transition, then it descends to a well-defined ledger-licensed morphism*

$$T_{\#} : \mathcal{P}_{\text{tar}} / \sim_{\text{pers}} \rightarrow \mathcal{P}_{\text{tar}} / \sim_{\text{pers}} .$$

If T maps $[p]_{\text{persist}}$ to a non-proton final class while no such transition is in the target ledger, then T is not an admissible same-target morphism. If the ledger is enlarged to include T , the target occupant has changed.

Proof. By the transition descent criterion, same-target admissibility forces a well-defined quotient action $T_{\#}$. By the ledger-licensing lemma, the quotient action must be represented in $\mathcal{T}_{\text{tar}}$ with witness, history, role, and boundary status. There are three exhaustive cases.

First, $T_{\#} \in \mathcal{T}_{\text{tar}}$. Then it is an admitted target transition. Since the unique target occupant contains no admitted $\Delta B_{\text{role}} \neq 0$ proton-decay morphism, this case is not a proton-decay role in SM_{book} .

Second, $T_{\#} \notin \mathcal{T}_{\text{tar}}$. Then applying it as a same-target transition uses an unlicensed quotient morphism. The expression may be syntactically writable, but it is not an admissible target transition.

Third, one adds $T_{\#}$ to $\mathcal{T}_{\text{tar}}$. This changes $\mathcal{O}_{\text{tar}}$, W_T , H_T , ρ_T , or E_T . Hence the occupant tuple has changed in a standing-bearing component. The result is not the same target occupant. These cases exhaust all same-target uses of T . $\square$

Corollary 8.7 (No same-target persistence-class destruction without ledger motion). *A claimed transition that destroys $[p]_{\text{persist}}$ while preserving all target transition ledgers is impossible. If it preserves all ledgers, it is not destruction of the proton persistence class; if it destroys the class, the ledger must contain or acquire the corresponding morphism.*

Proof. If the class is destroyed, the quotient action sends $[p]_{\text{persist}}$ to a non-proton class. By the theorem, such an action must either already be ledger-licensed or require ledger expansion. If it is neither, it is not an admissible same-target transition. $\square$

Remark 8.8 (Why the theorem is not tautological). The theorem does not merely say that changing a ledger changes a ledger. It proves that same-target transition claims are morphisms on the standing quotient and that morphisms on the quotient require ledger licensing. Proton-decay syntax fails not because a convention forbids it, but because a persistence-class destroying morphism is absent from the target ledger and cannot be inserted without changing the occupant.

9 Event projection and the Induced Target Transition Lemma

Transition-ledger rigidity blocks same-target insertion. A larger-domain theory may still contain a parent-domain process that it calls proton decay. The target question is not whether the parent process can be written. The target question is whether that parent event projects to a boundary-detectable loss of the target proton persistence class. If it does, the target transition witness is induced. If it does not, the parent event is not target proton decay.

9.1 Projection objects

Definition 9.1 (Operational target projection). Let L^+ be a larger-domain theory or completion and let e be a claimed event in L^+ . The target projection is the four-component package

$$\pi_{\text{tar}}(L^+, e) = (\partial_{\text{in}}, \partial_{\text{out}}, W_{\text{changed}}, \mathcal{T}_{\text{tar}}^{\text{needed}}),$$

where ∂_{in} is the target boundary record of the initial class, ∂_{out} is the target boundary record of the projected final family, W_{changed} is the set of target witnesses changed by the event, and $\mathcal{T}_{\text{tar}}^{\text{needed}}$ is the transition-ledger entry required to make the projected event admissible as target standing.

Definition 9.2 (Target boundary record). The target boundary record of a persistence class is written $\partial_{\text{tar}}([X]_{\text{persist}})$. For a claimed target proton-decay event the input record is

$$\partial_{\text{in}} = \partial_{\text{tar}}([p]_{\text{persist}}),$$

and the projected output record is $\partial_{\text{out}} = \partial_{\text{tar}}(F)$. A projected event is witness-bearing only when the target boundary distinguishes ∂_{in} from ∂_{out} under admissible identity-preserving comparison.

Definition 9.3 (Operational target statuses). The projection $\pi_{\text{tar}}(L^+, e)$ has one of three target statuses:

- same target quotient $W_{\text{changed}} = \emptyset$ and $\partial_{\text{out}} \simeq_{\text{Adm}} \partial_{\text{in}}$,
- induced target witness $W_{\text{changed}} \neq \emptyset$ and $\mathcal{T}_{\text{tar}}^{\text{needed}}$ is required,
- standing failure $\partial_{\text{out}} \not\simeq_{\text{Adm}} \partial_{\text{in}}$ is asserted without W_{changed} .

The first status licenses no target proton decay. The second status moves the target transition ledger. The third status carries no target standing.

Definition 9.4 (Event-level target decay). A parent-domain event e is a target proton-decay event only if its projection satisfies all three conditions:

1. $\partial_{\text{in}} = \partial_{\text{tar}}([p]_{\text{persist}})$ is defined;
2. ∂_{out} lacks the proton role package P_p ;
3. W_{changed} contains a target witness for the before/after difference.

If any condition fails, the parent-domain event may be meaningful in L^+ , but it is not an admitted target proton-decay event.

9.2 Event-projection witness lemma

Theorem 9.5 (Event-Projection Witness Lemma). *Let L^+ claim an actual proton-decay event e for the target proton persistence class $[p]_{\text{persist}}$. Then*

$$\pi_{\text{tar}}(L^+, e) \in \begin{cases} \text{same target quotient / no target decay,} \\ \text{target transition-witness induction,} \\ \text{standing failure.} \end{cases}$$

If e is genuinely a target proton-decay event, then

$$W_{\text{changed}} \neq \emptyset, \quad \partial_{\text{tar}}([p]_{\text{persist}}) \not\simeq_{\text{Adm}} \partial_{\text{tar}}(F),$$

and $\mathcal{T}_{\text{tar}}^{\text{needed}}$ contains or requires a ΔB_{role} transition morphism.

Proof. The four-component projection package makes the alternatives explicit. If $W_{\text{changed}} = \emptyset$, then no target witness distinguishes the input record from the output record. The event is either not represented at the target boundary or is same-quotient target persistence. It does not license target proton decay.

If $W_{\text{changed}} \neq \emptyset$ and ∂_{out} lacks P_p , then the target boundary distinguishes $\partial_{\text{tar}}([p]_{\text{persist}})$ from $\partial_{\text{tar}}(F)$. Because F lacks the proton role package and fails the baryonic witness package W_B , the changed witness set includes a ΔB_{role} transition witness. The projection therefore

requires a ledger entry $\mathcal{T}_{\text{tar}}^{\text{needed}}$ whose quotient action destroys the target proton persistence class.

If the event asserts $\partial_{\text{out}} \not\sim_{\text{Adm}} \partial_{\text{in}}$ while W_{changed} is empty or absent, the event has no target standing: an identity-bearing before/after difference has been claimed without the witness required to compare the records. These cases exhaust the projection because the changed-witness set is either empty, nonempty, or missing while a change is asserted. $\square$

9.3 Induced target transition

Theorem 9.6 (Induced Target Transition Lemma). *Let L^+ be any larger-domain theory, completion, or parent carrier that claims actual proton decay for the target proton persistence class. Then exactly one of the following target statuses holds:*

1. L^+ induces no standing-bearing target transition witness; then it does not license proton decay at the target level.
2. L^+ induces a standing-bearing $\Delta B_{\text{role}} \neq 0$ transition witness; then it changes the target transition ledger and is not a same-target occupant.
3. L^+ asserts target decay without a target witness; then it is hidden selector or standing failure.

Proof. Apply the Event-Projection Witness Lemma to $\pi_{\text{tar}}(L^+, e)$. If $W_{\text{changed}} = \emptyset$, case (1) holds. If $W_{\text{changed}} \neq \emptyset$ and the output lacks P_p , the target transition ledger must contain or acquire the corresponding quotient morphism $\mathcal{T}_{\text{tar}}^{\text{needed}}$; this is case (2). If the parent claim asserts the loss of $[p]_{\text{persist}}$ without any target witness, the transition is not target-standing; this is case (3). Since every target projection has exactly one changed-witness status, the three cases are exhaustive. $\square$

9.4 Parent-target commuting diagram

The larger-domain route can be summarized by the diagram

$$\begin{array}{ccc}
 L^+ & \xrightarrow{\text{parent transition}} & e_{p \rightarrow F}^+ \\
 \downarrow \pi_{\text{tar}} & & \downarrow \pi_{\text{tar}} \\
 \text{SM}_{\text{book}} & \xrightarrow{\text{target transition?}} & (\partial_{\text{in}}, \partial_{\text{out}}, W_{\text{changed}}, \mathcal{T}_{\text{tar}}^{\text{needed}}).
 \end{array}$$

The lower arrow is the whole issue. If it exists as $\Delta B_{\text{role}} \neq 0$, the target ledger moves. If it does not exist, the parent event does not license target proton decay. Thus a larger-domain event cannot be both target proton decay and invisible to the target transition ledger.

Corollary 9.7 (Same quotient or induced witness). *For any larger-domain proposal restricted*

to the target interface,

$$L^+|_{\text{tar}} = \begin{cases} \text{SM}_{\text{book}}, & \text{same quotient; no target proton-decay role,} \\ \text{SM}_{\text{book}} + W_T^{\Delta B}, & \text{induced target witness; target ledger motion,} \\ \perp, & \text{standing failure.} \end{cases}$$

Proof. If the restriction preserves the standing quotient, witnesses, transports, histories, role typing, and boundary ledger, it is the same target occupant and contains no proton-decay role. If it contains a standing-bearing ΔB_{role} witness, then $W_{\text{changed}} \neq \emptyset$ and the transition ledger differs. If it neither preserves the target quotient nor supplies the witness, it has no target standing. $\square$

10 Gauge-invariant B-violating EFT operators and transport standing

The standard objection is immediate: one can write gauge-invariant dimension-six operators schematically of the form

$$\mathcal{O}_{\Delta B}^{(6)} \sim \frac{qqql}{\Lambda^2}$$

that mediate baryon-violating transitions in ordinary effective-field-theory language. The AASC answer is not that EFT is illegitimate. The answer is that gauge-invariant syntax is not the same object as target transition standing. An EFT operator is admissible when its parent, matching, regime, error envelope, and target projection are carried. It is not admissible as a same-occupant primitive that silently creates a proton-decay role.

10.1 Transport diagram

The relevant standing diagram is

$$L^+ \xrightarrow{\text{match}} \mathcal{O}_{\Delta B}^{(6)} \xrightarrow{\pi_{\text{tar}}} T_{p \rightarrow F}^{\Delta B}.$$

The first arrow is parent-to-EFT transport. The second arrow is target projection. Both must carry standing. A formal operator expression alone supplies neither arrow. Gauge invariance checks compatibility with the gauge/charge syntax; it does not certify the operator as a transition morphism on target persistence classes.

10.2 Boxed EFT trichotomy

For a baryon-violating EFT operator $\mathcal{O}_{\Delta B}^{(6)}$, exactly one target reading is available:

1. **EFT as calculational or predictive skin.** The operator is a matched representation of a parent-domain carrier with a regime, error envelope, coefficient status, and transport certificate. It is admissible as larger-domain skin, not as a same-target primitive.
2. **EFT as target-domain transition witness.** The operator projects to a target ΔB_{role} transition witness. Then $W_{\text{changed}} \neq \emptyset$ and the target transition ledger moves.
3. **EFT as primitive same-occupant operator.** The operator is inserted directly into SM_{book} as a new transition term. Then it is new operator load, not redescription.

No reading yields a same-target admissible proton-decay role.

Definition 10.1 (EFT predictive skin). The operator $\mathcal{O}_{\Delta B}^{(6)}$ is an EFT predictive skin when it is a derived representation of a parent carrier, with matching rule, scale regime, error envelope, and transport certificate. In this reading it is not a same-target primitive. It is a skin of a larger-domain construction restricted to an admitted regime.

Definition 10.2 (Primitive same-target operator). The operator is primitive same-target load when inserted into the target transition/operator ledger without being derived by an admitted parent transport. In that reading it changes $\mathcal{O}_{\text{tar}}$, W_T , and the transition ledger.

Definition 10.3 (Witness-inert notation). The operator is witness-inert notation when it induces no target boundary-record change and no target quotient morphism. It may be written syntactically, but it licenses no target proton decay.

10.3 Branch lemmas

Lemma 10.4 (Gauge invariance is not transition standing). *Gauge invariance of $\mathcal{O}_{\Delta B}^{(6)}$ is not sufficient for AASC transition standing.*

Proof. Gauge invariance is a compatibility test under the gauge/charge syntax. AASC transition standing additionally requires a quotient action on target persistence classes, a transition witness, transport completeness, admissible histories, boundary record, and no-repair lineage. These are not supplied by gauge invariance alone. A gauge-invariant expression can therefore be a formal operator candidate while still lacking target transition standing. $\square$

Lemma 10.5 (Primitive baryon-violating EFT operator is new load). *If $\mathcal{O}_{\Delta B}^{(6)}$ is inserted as an independent target primitive, then the target operator and transition ledgers change.*

Proof. The target occupant did not contain a ΔB_{role} proton-decay morphism. Adding a primitive operator whose intended action is $[p]_{\text{persist}} \mapsto F$ adds an operator carrier and a transition witness. This changes $\mathcal{O}_{\text{tar}}$, W_T , and $\mathcal{T}_{\text{tar}}$. It is therefore new load, not same-target redescription. $\square$

Lemma 10.6 (Transport-image EFT requires an admissible parent). *If $\mathcal{O}_{\Delta B}^{(6)}$ is claimed as an EFT transport image, then the parent domain must supply a transport-complete matching certificate, including regime, scale, error envelope, parent carrier, coefficient status, and target projection.*

Proof. Without the parent/matching/error ledger, the EFT coefficient is a bare insertion. With the ledger, it is not a same-target primitive but a parent-domain image. The target question then shifts to the projection of the parent event: if it induces a target ΔB_{role} witness, the target ledger moves; if it does not, no target decay is licensed. This is precisely the target-projection trichotomy. $\square$

Theorem 10.7 (EFT Branch Exhaustion for B-Violating Operators). *A gauge-invariant baryon-violating EFT operator has exactly one of three target statuses:*

1. *admissible transport image of a larger-domain carrier;*
2. *primitive new target operator load;*
3. *witness-inert notation.*

In case (1), any target proton-decay claim triggers the Induced Target Transition Lemma. In case (2), the operator is new load. In case (3), no proton decay is licensed. No branch yields a same-target admissible proton-decay role.

Proof. If the operator is derived from a parent with a transport certificate, it is a transport image and is classified by target projection. If it is not derived but is inserted into the target ledger, it changes the operator/transition witness family and is new load. If it is neither derived nor ledger-active, it does no target witness work and licenses no decay. The three cases exhaust the standing statuses of the operator. $\square$

Remark 10.8 (EFT is not banned). The result is not an anti-EFT statement. EFT is admissible as approximate predictive skin when the parent, matching, and error envelope are carried. The forbidden move is treating a baryon-violating EFT operator as a same-target primitive while also claiming target proton decay.

11 Heavy mediators, GUT-type carriers, quantum-gravity carriers, and topological routes

The previous section handled the EFT expression. This section handles common parent-domain routes: heavy mediators or GUT carriers, quantum-gravity or nonlocal carriers, and

topological or sphaleron-type routes. The purpose is not to dismiss those programs. The purpose is to type their target-domain projection.

11.1 Heavy-mediator and GUT-type carriers

Definition 11.1 (Heavy mediator carrier). A heavy-mediator or GUT-type proton-decay carrier is written

$$X_{\Delta B} = (R_X, W_X, T_X, \partial_X),$$

where R_X is the parent representation/mediator role, W_X its parent witness package, T_X its transport/matching data, and ∂_X its target boundary projection data. Its induced target operator, if any, is

$$\text{Ind}_{\text{tar}}(X_{\Delta B}) = O_{\Delta B}^{\text{tar}}.$$

Definition 11.2 (Heavy-mediator target branch). The target branch of $X_{\Delta B}$ is the status of $\text{Ind}_{\text{tar}}(X_{\Delta B})$ after target projection. It is evaluated by whether the induced object is witness-inert, a target ΔB_{role} transition witness, a change of product/representation/charge/anomaly/vacuum/continuation ledger, or an uncertified assertion.

Theorem 11.3 (Heavy-Mediator Branch Exhaustion). *A heavy-mediator or GUT-type carrier $X_{\Delta B}$ has one of four target statuses:*

1. $\text{Ind}_{\text{tar}}(X_{\Delta B})$ is empty or witness-inert: no target proton decay is licensed.
2. $\text{Ind}_{\text{tar}}(X_{\Delta B})$ produces a ΔB_{role} transition witness: the target transition ledger moves.
3. $X_{\Delta B}$ changes charge, representation, anomaly, gauge, vacuum, or continuation ledgers: it is larger-domain load.
4. $X_{\Delta B}$ is asserted without transport certificate: it is standing failure.

No branch yields a same-target admissible proton-decay role.

Proof. If no target witness is induced, the target proton class persists or no target event exists. If a ΔB_{role} target witness is induced, the Induced Target Transition Lemma classifies it as target ledger motion. If the mediator requires new representation, charge, anomaly, product, vacuum, or continuation structure, it has moved the parent/target envelope and is not a same-target occupant. If no transport certificate exists, the parent claim does not descend to target standing. The four cases exhaust the target statuses of a mediator carrier. $\square$

Remark 11.4 (Why GUT mediation is not ignored). A GUT-style parent may be mathematically and phenomenologically meaningful. The theorem says only that, when restricted to the target proton-decay question, the parent either induces no target decay, induces a target witness and thus moves the target ledger, or lacks an admissible projection. This is a typing theorem, not a ban on studying the parent.

11.2 Quantum-gravity or nonlocal boundary carriers

A quantum-gravity or nonlocal proton-decay proposal is treated as a parent-domain carrier $\text{QG}_{\Delta B}$ with a target projection. No detailed quantum-gravity model is assumed or rejected. The same operational projection trichotomy applies:

$$\pi_{\text{tar}}(\text{QG}_{\Delta B}, e) = (\partial_{\text{in}}, \partial_{\text{out}}, W_{\text{changed}}, \mathcal{T}_{\text{tar}}^{\text{needed}}),$$

with status

$$\pi_{\text{tar}}(\text{QG}_{\Delta B}, e) \in \begin{cases} \text{same quotient,} \\ \text{induced target witness,} \\ \text{standing failure.} \end{cases}$$

Corollary 11.5 (Quantum-gravity route typing). *A quantum-gravity or nonlocal proposal licenses target proton decay only if its target projection has $W_{\text{changed}} \neq \emptyset$ and includes a boundary-detectable ΔB_{role} transition witness for $[p]_{\text{persist}}$. If it does, the target transition ledger moves. If $W_{\text{changed}} = \emptyset$, no target proton decay is licensed. If it asserts the event without a witness package, it fails standing.*

Proof. This is the Induced Target Transition Lemma with $L^+ = \text{QG}_{\Delta B}$. A parent-domain claim that never changes the target boundary record has $W_{\text{changed}} = \emptyset$ and is not target proton decay. A parent-domain claim that changes the target boundary record has $W_{\text{changed}} \neq \emptyset$; if the final target family lacks P_p , this is a ΔB_{role} witness and therefore target ledger motion. A parent-domain claim that asserts the change without witness support fails standing. $\square$

11.3 Topological and sphaleron-type routes

The theorem target is ordinary loss of the individual target proton persistence class. Topological or sphaleron-type bookkeeping may belong to a different admitted question. The present theorem does not classify every anomalous, thermal, instanton-like, or finite-temperature topological transition. It classifies target proton decay.

Theorem 11.6 (Topological Route Boundary). *Topological or sphaleron-type bookkeeping is outside the ordinary target proton-decay theorem unless it projects to a boundary-detectable loss of the individual target proton persistence class. If it does project that way, it induces a target transition witness and falls under the Induced Target Transition Lemma. If it does not, it is not a proton-decay carrier.*

Proof. A topological process that does not destroy $[p]_{\text{persist}}$ is not a target proton-decay event. If it does destroy $[p]_{\text{persist}}$, the target projection changes the boundary record and the B_{role} witness package. That is induced target transition-witness motion. Therefore the same target trichotomy applies. $\square$

12 Same quotient or induced witness

The larger-domain analysis can be compressed into a normal form. For any larger-domain proposal L^+ , its restriction or projection to the target occupant has one of three statuses:

$$L^+|_{\text{tar}} = \begin{cases} \text{SM}_{\text{book}}, & \text{same quotient,} \\ \text{SM}_{\text{book}} + W_T^{\Delta B}, & \text{induced target witness,} \\ \perp, & \text{standing failure or no target projection.} \end{cases}$$

Theorem 12.1 (Same Quotient or Induced Witness). *Let L^+ be a larger-domain theory or completion used to claim proton decay. If $L^+|_{\text{tar}} \simeq_{\text{Adm}} \text{SM}_{\text{book}}$, then no target proton-decay role is licensed. If $L^+|_{\text{tar}}$ contains a standing-bearing ΔB_{role} transition witness, the target transition ledger differs from SM_{book} . If neither condition holds, the claim has no target standing.*

Proof. Same quotient means the target standing quotient, witnesses, transports, histories, role typing, and boundary ledger agree with SM_{book} . Since SM_{book} has no admitted proton-decay quotient morphism, no target proton-decay role is present. If a ΔB_{role} transition witness is present, then by the Baryonic Role Standing theorem and Quotient-Morphism Transition Rigidity the target transition ledger differs. If the proposal neither preserves the quotient nor supplies the witness, it is not a standing-bearing target claim. $\square$

Remark 12.2 (Why this is not anti-BSM). The theorem does not say that larger-domain physics is impossible. It says that a larger-domain proposal has to be typed. Same-quotient proposals do not license target proton decay; induced-witness proposals move the target ledger; witness-free proposals do not carry target standing. This is exactly the BSM non-competition posture of the occupant classification.

13 Same-target proton stability

The same-target theorem now follows. It is not the full occupant-level theorem, because it assumes the target occupant is fixed. It proves that no proton-decay morphism can be introduced while preserving that occupant.

Theorem 13.1 (Same-Target Proton Stability). *Inside SM_{book} , there is no admissible same-target transition morphism*

$$T_{p \rightarrow F}^{\text{Adm}} : [p]_{\text{persist}} \longrightarrow F$$

with $\Delta B_{\text{role}}(T_{p \rightarrow F}^{\text{Adm}}) \neq 0$.

Proof. Assume such a transition exists. By Quotient-Morphism Transition Rigidity, it must descend to a ledger-licensed morphism $T_{\#}$ on $\mathcal{P}_{\text{tar}}/\sim_{\text{pers}}$. Because $\Delta B_{\text{role}} \neq 0$, the transition changes the witnessed baryonic persistence role and therefore changes the target transition

ledger unless it is already admitted. But the target transition ledger of SM_{book} contains no admitted proton-decay quotient morphism. If $T_{\#}$ is not in the ledger, the transition is unlicensed. If it is added, the occupant changes. Thus no same-target admissible proton-decay transition exists. $\square$

Corollary 13.2 (No internal lifetime parameter). *Because the target decay channel is not admitted, there is no internal proton-lifetime parameter in SM_{book} . A lifetime parameter would be meaningful only after a ΔB_{role} transition role were admitted.*

14 Redescription exhaustion

A proton-decay role cannot be generated by a skin-level change. The following route table records the executive exhaustion; Appendix M gives the full audit version.

Route	Standing effect	Disposition
Gauge redescription	Preserves target standing classes	No decay
Basis/flavor redescription	Changes coordinates or labels only	Skin
Charge normalization/sign	Normalization or orientation convention	No transition
Finite parameter correction	Can alter a rate only after a channel exists	Cannot create channel
Mass/Yukawa insertion	Fills mass-response or coupling slots	No $O_{\Delta B}$
Primitive $qqql/\Lambda^2$	Adds transition operator ledger	New load
Transport-image $qqql/\Lambda^2$	Requires parent and target projection	Induced-target trichotomy
Heavy GUT mediator	New target witness or none	Ledger motion or no decay
Quantum-gravity carrier	New target witness or none	Ledger motion or no decay
Topological route	Target proton loss or not	Induced-target trichotomy
Hidden decay label	Unwitnessed distinction	Standing failure
Post-selection repair	Retroactive standing assignment	Inadmissible repair
Empirical event claim	Boundary-record audit required	Dependency audit or target ledger motion

Theorem 14.1 (Redescription route exhaustion). *No admissible redescription of the target*

occupant produces a proton-decay transition role. Every route is skin, no target decay, new load, target ledger motion, or standing failure.

Proof. A redescription preserves the standing quotient, witness family, transport/history structure, and role typing. If a route preserves these data, it cannot destroy $[p]_{\text{persist}}$. If it destroys $[p]_{\text{persist}}$, it changes the transition ledger or requires a non-licensed morphism. The listed routes exhaust the standard ways of attempting to smuggle a transition into the target occupant. None yields a same-target proton-decay role. $\square$

15 Charge, anomaly, and transition-ledger constraints

The theorem does not rest on electric charge conservation alone. Channels such as $p \rightarrow e^+ \pi^0$ can preserve electric charge. The obstruction is not that the final state fails a charge sum. The obstruction is that a persistence-destroying ΔB_{role} transition morphism is not a target-ledger morphism.

Lemma 15.1 (Charge neutrality is not transition admissibility). *A charge-neutral or gauge-invariant expression is not automatically an admissible transition carrier. Operator standing requires target-ledger admission.*

Proof. Charge or gauge compatibility checks a subset of the structural ledgers. A transition carrier also requires persistence-domain action, quotient morphism, witness package, history compatibility, boundary record, and no-repair lineage. Therefore neutrality is not sufficient for transition standing. $\square$

Proposition 15.2 (Use of charge-lattice rigidity). *Charge-lattice rigidity blocks attempts to obtain proton decay through charge-normalization, additive-offset, hidden-selector, extra-abelian, or exotic same-scope charge routes. It does not by itself prove proton stability; it supplies a local route-exhaustion pattern.*

Proof. The charge-lattice papers show that continuous relative charge deformation, non-lattice interpolation, additive offset, hidden charge selector, extra $U(1)$, extra chiral matter, and additional Higgs multiplet are same quotient, inadmissible, or larger scope in fixed charge reconstruction [5, 6]. The proton-decay theorem uses the same AASC typing pattern at the transition-ledger level. Charge balance may be satisfied, but transition admission remains absent. $\square$

16 From same-target stability to occupant-level exclusion

The same-target theorem blocks proton decay inside SM_{book} . The occupant-level theorem uses the Classification Theorem to block a non-equivalent target occupant obtained by adding such a transition role.

Definition 16.1 (Hypothetical $+\Delta B$ target occupant). Let

$$\text{SM}_{\text{book}}^{+\Delta B} = \text{SM}_{\text{book}} + T_{p \rightarrow F}^{\Delta B}$$

denote a hypothetical target occupant obtained by adding a standing-bearing proton-decay transition morphism.

Lemma 16.2 (Non-equivalence of a standing $+\Delta B$ extension). *If $\text{SM}_{\text{book}}^{+\Delta B}$ contains a standing-bearing ΔB_{role} proton-decay transition, then*

$$\text{SM}_{\text{book}}^{+\Delta B} \not\simeq_{\text{Adm}} \text{SM}_{\text{book}}.$$

Proof. If the added transition changes no target witness, it is skin and not proton decay. If it is standing-bearing, it changes W_T , $\mathcal{T}_{\text{tar}}$, the persistence-continuation structure, or the boundary record. Occupant equivalence requires preservation of witnesses, quotient, transport/history structure, role typing, and boundary ledger. Therefore a standing $+\Delta B$ extension is non-equivalent. $\square$

Theorem 16.3 (Occupant-Level Proton Stability). *Assume the AASC Classification Theorem:*

$$B_{\text{AASC}} / \simeq_{\text{Adm}} = \{[\text{SM}_{\text{book}}]\}.$$

Then there is no admissible target particle-physics occupant containing a standing-bearing proton-decay transition role.

Proof. If such an occupant exists and is equivalent to SM_{book} , the transition must preserve the same quotient and transition ledger; by same-target proton stability, it is not proton decay. If it is not equivalent, it contradicts occupant uniqueness. If it is asserted without target standing, it is not an admissible occupant. Thus no admissible target occupant contains the proton-decay role. $\square$

17 Standing conservation and terminal normal form

The previous theorem may be read in categorical normal-form language. This is not extra ontology; it is an audit compression of occupant uniqueness.

Definition 17.1 (Standing flux). For a fixed target scope, a morphism has zero standing flux when it creates no standing class, destroys no standing class, and changes no invariant witness ledger except by admissible isomorphism.

Proposition 17.2 (Proton decay has nonzero standing flux). *A transition $T_{p \rightarrow F}$ with $\Delta B_{\text{role}} \neq 0$ has nonzero standing flux unless it is already an admitted ledger morphism.*

Proof. It destroys the witnessed baryonic persistence-role content of $[p]_{\text{persist}}$ and maps the class to a final family lacking P_p . That changes standing content unless the transition is already represented in the transition ledger. Since no such morphism is present in SM_{book} , the proposed transition has nonzero standing flux relative to the target occupant. $\square$

Theorem 17.3 (Terminal-normal-form reading). *In the target particle-physics component of the AASC normal-form category, SM_{book} is the terminal normal form. A proton-decay extension would either factor through this normal form without decay or require a non-terminal standing-bearing extension.*

Proof. This restates the Classification Theorem’s terminal-object content in the proton-decay target. If the extension factors through SM_{book} , it has the same transition ledger and no proton-decay morphism. If it does not factor through, it is not the unique target normal form. $\square$

18 Main theorem package

This section gathers the proof chain into discrete theorem-level obligations. Each theorem discharges one bridge. No single theorem silently combines proton admission, baryonic-role witnessing, transition-ledger rigidity, larger-domain projection, and occupant uniqueness.

18.1 Role admission and baryonic witnessing

Theorem 18.1 (Proton role admission). *The admitted proton persistence role $[p]_{\text{persist}}$ is a target standing class under the Proton Persistence Certificate and the Hadronic Persistence Admission Lemma.*

Proof. The certificate fixes localized persistence, color-neutral composite role, charge-lattice presentation, mass response, boundary detectability, and baryonic role witness status. The hadronic admission lemma states that such color-neutral composite persistence classes are admitted downstream of the recovered target occupant when their admission datum is fixed. Therefore $[p]_{\text{persist}}$ is a legitimate target standing class for the theorem. $\square$

Theorem 18.2 (Baryonic role witnessing). *B_{role} is witness-defined by W_B , not primitive.*

Proof. By the Baryonic Role Standing theorem, B_{role} has target standing only through the witness package W_B . The classification-first independence result shows that W_B records baryonic composite role standing rather than a no-decay axiom. A witness-free B -label is skin or hidden selector failure. Therefore the theorem does not rely on primitive baryon-number conservation. $\square$

Theorem 18.3 (Baryonic-role loss is transition-ledger motion).

$$\Delta B_{\text{role}} \neq 0 \Rightarrow \Delta \mathcal{T}_{\text{tar}} \neq 0$$

unless the transition is already admitted as a ledger-licensed quotient morphism.

Proof. Loss of B_{role} changes at least one component of W_B , hence changes the persistence-continuation witness ledger. The transition ledger is the admissibility structure for such changes. If the ledger already contains the morphism, the transition is admitted; otherwise adding it moves the ledger. $\square$

18.2 Same-target transition rigidity

Theorem 18.4 (Quotient-morphism transition rigidity). *A same-target admissible transition must descend to a ledger-licensed quotient morphism on $\mathcal{P}_{\text{tar}}/\sim_{\text{pers}}$.*

Proof. Representative-dependent transitions cannot preserve standing under irreversible reuse. Therefore any same-target transition descends to $T_{\#}$. A bare quotient function is not enough; it must be represented in $\mathcal{T}_{\text{tar}}$ with witness, history, role, and boundary status. This is exactly the Quotient-Morphism Transition Rigidity theorem of Section 8. $\square$

Theorem 18.5 (Same-target proton stability).

$$\text{SM}_{\text{book}} \vdash \neg \exists T_{p \rightarrow F}^{\text{Adm, same}} (\Delta B_{\text{role}} \neq 0).$$

Proof. Assume such a transition exists. By quotient-morphism transition rigidity, it must descend to a ledger-licensed morphism $T_{\#}$ on $\mathcal{P}_{\text{tar}}/\sim_{\text{pers}}$. Because $\Delta B_{\text{role}} \neq 0$, the transition changes the witnessed baryonic persistence role and therefore changes the target transition ledger unless it is already admitted. But the target transition ledger of SM_{book} contains no admitted proton-decay quotient morphism. If $T_{\#}$ is not in the ledger, the transition is unlicensed. If it is added, the occupant changes. Thus no same-target admissible proton-decay transition exists. $\square$

18.3 Larger-domain projection

Theorem 18.6 (Induced target transition exhaustion). *Any larger-domain proton-decay claim either induces target ledger motion, licenses no target decay, or fails standing.*

Proof. Let L^+ claim proton decay by a parent event e . Apply the operational event projection

$$\pi_{\text{tar}}(L^+, e) = (\partial_{\text{in}}, \partial_{\text{out}}, W_{\text{changed}}, \mathcal{T}_{\text{tar}}^{\text{needed}}).$$

If $W_{\text{changed}} = \emptyset$, no target proton decay is licensed. If $W_{\text{changed}} \neq \emptyset$ and the output lacks P_p , ΔB_{role} transition-ledger motion occurs. If the event asserts destruction without target witness content, the claim fails standing. This is the Induced Target Transition Lemma. $\square$

Corollary 18.7 (EFT and mediator routes). *Gauge-invariant baryon-violating EFT terms, heavy mediators, GUT carriers, quantum-gravity carriers, and topological proposals do not define same-target proton-decay roles. They are same-quotient/no-decay, induced-witness/target-ledger-motion, or standing failure.*

Proof. EFT operators are classified by EFT branch exhaustion; heavy mediators by heavy-mediator branch exhaustion; quantum-gravity and topological routes by target projection. Each route reduces to the same quotient or induced witness trichotomy. $\square$

18.4 Occupant-level theorem

Theorem 18.8 (Occupant-level proton stability). *Inside the unique target AASC particle-physics occupant,*

$$\neg \exists T_{p \rightarrow F}^{\text{Adm}}(\Delta B_{\text{role}} \neq 0).$$

Proof. A same-target transition is blocked by same-target proton stability. A larger-domain transition either does not project to target decay, induces target ledger motion, or fails standing. Target ledger motion yields a non-equivalent target occupant because the transition/operator/witness/history/role ledger differs. The Classification Theorem gives

$$B_{\text{AASC}} / \simeq_{\text{Adm}} = \{[\text{SM}_{\text{book}}]\}.$$

Hence no admissible target occupant contains the decay role. $\square$

Corollary 18.9 (No internal proton-lifetime parameter). *The target occupant contains no internal proton-lifetime parameter for ordinary proton decay. A lifetime parameter can only be assigned after a decay channel is admitted. Since no target proton-decay transition role is admitted, there is no such internal parameter to compute.*

Proof. A rate parameter is a value inside an admitted transition channel. The theorem blocks the channel as a target transition role. Therefore the parameter slot is absent rather than numerically unknown inside the target occupant. $\square$

19 Relation to conventional proton-decay physics

This section records the conventional physics interface. It does not add a premise to the AASC proof. Its function is to make clear that the theorem is not competing with proton-lifetime phenomenology, GUT model building, or effective-field-theory practice. The AASC result is conditional on the locked AASC structural-recovery and occupant-uniqueness dependencies; its contribution is the transition-role exclusion derived from those dependencies.

19.1 Renormalizable Standard Model baseline

The conventional baseline is that ordinary proton decay is absent from the renormalizable Standard Model. In standard language, baryon number and lepton number are often described as accidental symmetries of the renormalizable perturbative Standard-Model operator content, while nonperturbative and topological $B + L$ effects are treated separately from ordinary

proton-decay channels; recent baryon/lepton-number reviews use this distinction to motivate both experimental and BSM searches [16, 17]. AASC agrees with the fixed-load baseline but uses a different diagnostic: it asks whether a proton-decay transition role can have standing in the unique target occupant.

Thus the AASC result is not the claim that physicists cannot write a baryon-violating expression. It is a claim about standing: a proton-decay expression has to become a target-domain transition witness before it can license decay of the target proton persistence class.

19.2 GUT motivation and heavy mediators

The historical motivation for proton decay is closely tied to grand unification. In the Georgi–Glashow SU(5) construction, strong, electromagnetic, and weak interactions are organized through a single unifying gauge group, creating the familiar route by which quark and lepton roles can be related in a larger representation structure [9]. Broad reviews of proton stability in grand-unified, string, and brane settings treat proton decay as a characteristic probe of larger-domain unification structures rather than as an ordinary parameter value inside the renormalizable Standard Model [14]. Supersymmetric GUT analyses likewise treat proton decay as a process mediated by additional high-scale structure [13].

In the present manuscript, a GUT-type mediator is therefore not dismissed. It is typed. If its target projection is same quotient, no target proton decay is licensed. If it induces a target-domain ΔB_{role} transition witness, it moves the target transition ledger. If it is asserted without transport and boundary witnesses, it fails standing. This is exactly the event-projection trichotomy developed above.

19.3 Effective operators and the Weinberg route

Effective-operator analyses are part of the standard proton-decay literature. Weinberg’s baryon- and lepton-nonconserving process analysis, Wilczek–Zee’s operator analysis of nucleon decay, and later operator-dimension classifications provide the usual language for baryon-violating processes [10–12, 15]. In conventional EFT practice, a higher-dimensional baryon-violating operator can be an appropriate low-energy representation of heavier physics, provided the matching, domain, and error envelope are part of the calculation.

AASC does not ban that practice. The theorem separates three readings. First, an EFT operator can be a calculational skin or transport image of a larger-domain parent. Second, it can be used as a target-domain transition witness, in which case the target ledger has changed. Third, it can be inserted as a primitive same-occupant operator, in which case it is new load. None of those readings yields a same-target admissible proton-decay role.

19.4 Experimental status and lifetime limits

Experimentally, proton decay remains unobserved. Super-Kamiokande searches in the canonical modes $p \rightarrow e^+\pi^0$ and $p \rightarrow \mu^+\pi^0$ used a 450 kton·yr exposure and set 90% confidence lower limits $\tau/B(p \rightarrow e^+\pi^0) > 2.4 \times 10^{34}$ yr and $\tau/B(p \rightarrow \mu^+\pi^0) > 1.6 \times 10^{34}$ yr [18]. The SUSY-GUT-motivated $p \rightarrow \nu K^+$ channel has been searched with 260 kton·yr of Super-Kamiokande data, yielding $\tau/B(p \rightarrow \nu K^+) > 5.9 \times 10^{33}$ yr at 90% confidence [19]. The Particle Data Group maintains the broader review and summary-table context for particle properties and search limits [20].

The AASC theorem does not compute any of these partial lifetimes and does not replace the searches. It classifies what a positive event would have to mean structurally. A reproducible proton-decay event would either fail to be a target proton-persistence event, induce a new target transition witness, expose a standing failure in the event description, or challenge a locked AASC dependency. That is an empirical interface, not an attempt to make experiment irrelevant.

19.5 Comparator summary

The conventional story says: proton decay is absent from the renormalizable Standard Model, is naturally motivated by many unification or BSM frameworks, can be represented in EFT by higher-dimensional baryon-violating operators, and remains experimentally unobserved. The AASC story says: once the target particle-physics occupant is fixed by the locked structural-recovery and occupant-uniqueness results, a proton-decay expression must be classified as same quotient, induced target witness, primitive new load, or standing failure. The proof concerns that classification. It is orthogonal to experimental lifetime bounds and model-specific GUT calculations.

20 Empirical interface and possible failure modes

Theorem 20.1 (Empirical Proton-Decay Claim Interface). *A reproducible empirical proton-decay claim has one of four AASC statuses:*

1. *it is not a target proton-persistence event;*
2. *it is a target event with a new transition witness, hence target ledger motion;*
3. *it is a same-target event preserving all ledgers, in which case it refutes Transition-Ledger Rigidity or occupant uniqueness;*
4. *it is a standing-failure or misclassification under the declared target interface.*

Proof. A boundary event either projects to the target proton persistence class or not. If it does not, it is not a target proton-decay event. If it does and the proton class is lost, the boundary record changes and a target transition witness is induced. If the event claims

loss while preserving all ledgers, it contradicts the quotient-morphism theorem. If the event cannot be given a standing-preserving projection, it is misclassification or standing failure relative to the target interface. $\square$

Remark 20.2 (Empirical accountability). This theorem is deliberately uncomfortable. A confirmed target proton-decay event would not be absorbed by calling it skin. It would either enlarge the target witness ledger or force failure of a locked AASC dependency. That is the correct empirical posture for a structural theorem.

21 Counterexample burden

A genuine same-target proton-decay counterexample must satisfy an inconsistent burden. This section states the burden as a theorem rather than as a checklist.

Theorem 21.1 (Proton-Decay Counterexample Burden Is Inconsistent). *A same-target counterexample must jointly provide:*

1. *an admitted proton persistence class $[p]_{\text{persist}}$;*
2. *a $\Delta B_{\text{role}} \neq 0$ transition witness;*
3. *preservation of the target standing quotient;*
4. *preservation of the target transition ledger;*
5. *no new operator load;*
6. *no new witness family;*
7. *no continuation/history change;*
8. *no hidden selector;*
9. *no repair or post-selection;*
10. *non-equivalence to skin.*

Items (2) and (4)–(7) cannot be jointly satisfied unless the alleged transition is already admitted. Since SM_{book} has no admitted proton-decay morphism, the burden is inconsistent.

Proof. A ΔB_{role} transition witness changes the witnessed baryonic persistence-role package W_B . If the target transition ledger is preserved, the witness must already be represented by a ledger-licensed quotient morphism. It is not. If the witness is added, then the transition ledger changes and at least one of the following also changes: the operator load, the witness family, the admissible history structure, or the role typing. If the alleged witness is not added but the transition is still asserted, the proposal is hidden selector or repair. Therefore the counterexample requirements cannot be jointly met. $\square$

Proposition 21.2 (Operational counterexample form). *A counterexample must exhibit a concrete object*

$$([p]_{\text{persist}}, T_{\#}, W_T^{\Delta B}, \mathcal{T}_{\text{tar}}, H_T, \rho_T, E_T)$$

together with a target projection package

$$\pi_{\text{tar}}(L^+, e) = (\partial_{\text{in}}, \partial_{\text{out}}, W_{\text{changed}}, \mathcal{T}_{\text{tar}}^{\text{needed}})$$

such that $W_{\text{changed}} \neq \emptyset$, ∂_{out} lacks P_p , every target occupant ledger is preserved, no new witness is introduced, and the result is not skin. These requirements are mutually inconsistent.

Proof. If $W_{\text{changed}} \neq \emptyset$, then some target witness changes. If ∂_{out} lacks P_p , the changed witness includes the baryonic persistence package W_B . A target transition with this witness must either already be in $\mathcal{T}_{\text{tar}}$, be added to $\mathcal{T}_{\text{tar}}$, or be asserted without a ledger entry. It is not already in $\mathcal{T}_{\text{tar}}(\text{SM}_{\text{book}})$. Adding it changes the ledger. Asserting it without addition fails standing. Hence no concrete object satisfying the displayed requirements exists inside the target occupant. $\square$

Counterexample claim	Required failure point
Gauge-invariant qqql/ Λ^2 is same-target	Must deny EFT Branch Exhaustion or prove the operator is already ledger-licensed.
GUT mediator is same-target	Must show induced target decay without target ledger motion.
Quantum gravity violates baryonic roles	Must supply target projection and show quotient preservation despite ΔB_{role} .
Observed event is same-target decay	Must refute the empirical interface theorem, quotient-morphism rigidity, or occupant uniqueness.
Hidden label selects decay-enabled proton	Must supply witness or abandon hidden-selector status.
Topological route is ordinary proton decay	Must project to individual target-proton loss and then avoid induced target witness motion.

Criterion 21.3 (Counterexample construction burden). *A successful counterexample must exhibit a concrete object*

$$([p]_{\text{persist}}, T_{\#}, W_T^{\Delta B}, \mathcal{T}_{\text{tar}}, H_T, \rho_T, E_T)$$

that destroys $[p]_{\text{persist}}$, preserves every target occupant ledger, adds no witness, and is not skin. The theorem says no such object can be constructed without denying one of the locked dependencies or leaving the target domain.

22 Referee objection map

The table records the principal objections and the exact theorem or definition a critic must challenge.

Attack	Required response
“You only proved the renormalizable SM lacks proton decay.”	Denied by occupant-level theorem and induced target projection. The proof is not merely a fixed-load Lagrangian check.
“The proton is assumed.”	The proton role is admitted by the Proton Persistence Certificate and Hadronic Persistence Admission Lemma; QCD numerics are expressly nonclaimed.
“You smuggle in baryon conservation.”	B_{role} is witness-defined by W_B , not primitive. Loss of the witnessed role is transition-ledger motion.
“Transition-Ledger Rigidity is tautological.”	The quotient-morphism proof shows same-target transitions must descend to ledger-licensed morphisms on standing classes.
“A larger-domain decay can occur while the target remains SM-like.”	Event projection says a target decay changes the target boundary record. Same quotient means no target decay; induced witness means target ledger motion.
“Gauge-invariant dimension-six operators exist.”	Gauge invariance is not transition standing. The EFT route is transport image, primitive new load, or witness-inert notation.
“GUTs predict proton decay.”	A GUT mediator either induces no target witness, induces a target ΔB_{role} witness, changes ledgers, or lacks transport certificate.
“Quantum gravity violates global symmetries.”	The proof does not rely on primitive global baryon symmetry. QG claims must project to target witnesses or not license target proton decay.
“Sphalerons violate baryon number.”	The theorem target is ordinary loss of the individual target proton persistence class. A topological route must project to such loss before the theorem applies.
“An observed event refutes the theorem automatically.”	An observed event is classified by the empirical interface theorem: target event with new witness, same-target contradiction, misclassification, or dependency failure.
“The result is unfalsifiable.”	Same-target observed proton decay preserving all ledgers would falsify transition-ledger rigidity or occupant uniqueness.
“This forbids BSM.”	No. BSM is typed as same quotient/no decay, induced witness/larger domain, or standing failure.

23 Transition-morphism audit dossier

The quotient-morphism theorem is the central referee-audit bridge. This section expands the proof obligation in operational terms. A target transition is not merely a symbol that takes one written state into another written state. It is an admissible operation on standing classes. That is why representative independence is the first audit test.

23.1 Representative independence

Suppose two target records $x, x' \in \mathcal{P}_{\text{tar}}$ are equivalent:

$$x \sim_{\text{pers}} x'.$$

This means that all target persistence witnesses, boundary records, admissible continuations, and role-typing facts relevant to the target scope identify them as the same persistence class. A transition rule T that gives inequivalent outputs for x and x' is not a physical difference discovered by the transition. It is a failure of the rule to respect the identity carrier on which the target occupant is built. Therefore an admissible transition must satisfy

$$x \sim_{\text{pers}} x' \implies T(x) \sim_{\text{pers}} T(x').$$

This is the exact analogue of admissible redescription in the static ledger. The difference is that the object being transported is not a charge label or coordinate chart, but a continuation of a persistence class.

23.2 Ledger license versus bare quotient function

Representative independence is necessary but still not sufficient. A bare function on the quotient may be mathematically well-defined while not being a licensed target transition. The target transition ledger specifies which quotient actions have standing as physical transitions. This matters for proton decay because the proposed rule does not merely permute or preserve an existing target class; it removes the proton role package and its baryonic persistence witnesses. The ledger license must therefore say not only that a quotient map exists, but that a morphism with this role-loss profile is admitted.

The critical objection often compresses these two levels. It says: “if the parent theory gives a transition, the target quotient can record the before and after states.” The correct AASC answer is: a before/after notation does not by itself produce a target morphism. The target morphism must be part of the standing-preserving transition ledger or must be induced as a new witness. In the latter case, the target ledger has changed.

23.3 Why destruction cannot be silent

A silent destruction of $[p]_{\text{persist}}$ would have to satisfy three conditions at once:

1. the boundary record distinguishes the pre-event proton class from the post-event final family;
2. the transition changes no target witness, quotient, role, or history ledger;
3. the transition is standing-bearing rather than notation.

These conditions are incompatible. If the boundary record distinguishes before and after, a target witness has changed. If no target witness changes, the claimed destruction is not a target standing fact. If the claimed destruction is retained without witnesses, it is hidden selector or repair. This is the concrete content of transition-ledger rigidity.

23.4 Audit criterion for a proposed transition

A proposed transition carrier T must pass the following audit:

1. identify the domain persistence classes;
2. identify the codomain persistence classes;
3. show representative independence on $\mathcal{P}_{\text{tar}}/\sim_{\text{pers}}$;
4. name the ledger entry licensing the quotient action;
5. state the transition witnesses;
6. state the boundary record and event projection;
7. identify whether any role, witness, history, operator, or envelope item changes;
8. if a parent domain is used, supply matching and transport data;
9. prove no hidden selector or post-hoc filter is used.

No proton-decay proposal satisfies this list while preserving the unique target occupant. If the list is satisfied by adding ΔB_{role} transition witnesses, the target ledger has moved. If it is not satisfied, the proposal has no target standing.

23.5 Comparison with ordinary scattering

The audit does not forbid ordinary interactions. In a scattering or dressing process, the proton persistence class may be represented differently across charts, momentum states, environmental contexts, or interaction histories while the class remains ledger-equivalent:

$$[p_{\text{in}}]_{\text{persist}} \simeq_{\text{Adm}} [p_{\text{out}}]_{\text{persist}}.$$

The quotient morphism preserves the proton role package. Proton decay is different because it asserts

$$[p]_{\text{persist}} \not\vdash_{\text{Adm}} F.$$

That inequivalence is exactly what forces the ledger question. Thus the theorem is not a ban on dynamical evolution. It is a ban on unlicensed persistence-class destruction inside a fixed target occupant.

24 Event-projection audit dossier

The event-projection theorem is the bridge that prevents “larger domain” from becoming an escape hatch. This section expands the operational meaning of target projection.

24.1 Parent event versus target event

A parent-domain theory L^+ may contain structures absent from SM_{book} : heavy mediators, unified representations, nonlocal carriers, boundary objects, or effective operators after matching. Such parent structures are not automatically target structures. To affect the target proton persistence class, a parent event e must have a target projection:

$$\pi_{\text{tar}}(L^+, e).$$

The projection is not assumed to be faithful to every parent detail. It asks one narrow question: what target boundary record, if any, is induced for $[p]_{\text{persist}}$? In the operational form used in the proof, the projection is the four-part package

$$\left(\partial_{\text{in}}, \partial_{\text{out}}, W_{\text{changed}}, \mathcal{T}_{\text{needed}}\right).$$

The first two entries identify the target before/after boundary records; the third records the target witnesses changed by the event; the fourth records the transition-ledger item needed to make the update admissible. This package gives the theorem an operational test: no changed target witness means no target proton-decay standing; a changed target witness means target-ledger motion.

24.2 Projection statuses

There are four raw projection statuses:

1. no target record is induced;
2. a target record is induced but preserves $[p]_{\text{persist}}$;
3. a target record is induced and destroys $[p]_{\text{persist}}$ with witnesses;
4. a target record is asserted to destroy $[p]_{\text{persist}}$ without witnesses.

The first two are not target proton decay. The third is target transition-witness induction. The fourth is standing failure. This is why any parent-domain decay claim collapses to the trichotomy used in the main theorem.

24.3 Boundary record as witness

The boundary record $\partial_{\text{tar}}([p]_{\text{persist}})$ is not a merely epistemic observation tag. It is the target interface by which the persistence class is reusable under AASC standing. If a projected event changes this boundary record from proton to non-proton final family, the target record itself supplies a distinguishing witness:

$$\partial_{\text{tar}}([p]_{\text{persist}}) \not\sim_{\text{Adm}} \partial_{\text{tar}}(F).$$

That witness is sufficient to move the target transition ledger because the transition ledger is the structure that determines which boundary-distinguishable persistence transitions are admissible.

24.4 Rare events do not avoid projection

A critical objection says that a rare parent-domain proton decay could occur while the target theory remains SM-like except at the moment of the rare event. The event-projection theorem answers: at the moment of the claimed event, either the target record remains proton-like, or it does not. If it remains proton-like, there is no target proton decay. If it does not, the target boundary record changes. Rarity affects frequency, not standing. A rare transition still needs a transition morphism.

24.5 Parent-domain truth and target-domain standing

The theorem does not deny that a parent-domain theory may be true in a larger sense. It says that truth in a larger domain does not by itself produce target standing. The parent event must be projected. If the projection induces a target ΔB_{role} witness, then the target occupant has been enlarged or superseded. If it does not, the target proton-decay claim has not been licensed.

24.6 Event projection and falsification

If a future experiment reported a reproducible proton-decay event, the first AASC audit question would not be “which familiar model predicted it?” The first question would be: what is the target projection of the boundary record? If the event is a confirmed loss of $[p]_{\text{persist}}$, then a target transition witness has been supplied. The target occupant theorem would then either be superseded by a larger-domain witness architecture or one of its locked dependencies would fail. This is a real empirical vulnerability, not a verbal escape.

25 EFT and GUT transport audit diagrams

This section gives the explicit transport diagrams behind the EFT and heavy-mediator branch theorems.

25.1 EFT diagram

The EFT path has three nodes:

$$L^+ \xrightarrow{\text{matching}} \mathcal{O}_{\Delta B}^{(6)} \xrightarrow{\text{target projection}} T_{p \rightarrow F}^{\Delta B}.$$

The first arrow is a parent-to-EFT transport. It requires a parent carrier, a regime declaration, an error envelope, and a matching rule. The second arrow is not automatic. It asks whether the EFT expression is being used merely as an approximate calculation device or as a target transition witness. If it is a calculation device, then it belongs to the larger-domain representation and does not by itself alter SM_{book} . If it is a transition witness, then it moves $\mathcal{T}_{\text{tar}}$.

25.2 Primitive-operator failure

If the operator is inserted directly as a target primitive, the diagram loses its first arrow. Then $\mathcal{O}_{\Delta B}^{(6)}$ has no parent transport certificate. It is simply a new operator-availability ledger entry. This is the cleanest new-load case. Its gauge invariance may make it syntactically compatible with part of the target gauge ledger, but syntactic compatibility is not same-occupant standing.

25.3 GUT mediator diagram

A GUT-type route has the diagram

$$X_{\Delta B} \xrightarrow{\text{integrate/match}} \mathcal{O}_{\Delta B}^{(6)} \xrightarrow{\pi_{\text{tar}}} T_{p \rightarrow F}^{\Delta B}.$$

The mediator $X_{\Delta B}$ may be a perfectly good parent-domain object. The target question is whether its target projection induces a new target transition witness. If the target projection is empty, no target proton decay is licensed. If the target projection is nonempty and decay-positive, the target ledger has moved. If the mediator changes representation, charge, anomaly, or product ledgers, it is larger-domain load even before the event projection is considered.

25.4 Why low-energy agreement is insufficient

A parent theory can reproduce all ordinary low-energy target outputs while also containing rare parent transitions. Ordinary output agreement is insufficient for target standing because AASC admissibility requires transport, continuation, quotient preservation, and boundary-record reuse. If the rare transition projects to a new target boundary record, then ordinary low-energy agreement before the event does not preserve the full target transition ledger. If it never projects, then it is not target proton decay.

25.5 No prohibition on transport skins

The audit does not ban EFT. It bans a specific conflation: treating a transported larger-domain operator as though it were simultaneously a same-target primitive and a target decay witness. A transported operator may be used for calculation if its parent and error envelope are carried. It becomes a target structural challenge only when it is used to license ΔB_{role} loss of $[p]_{\text{persist}}$.

26 Boundary witness and baryonic-role audit

This section expands the witness status of B_{role} . The purpose is to show that the proof does not define proton decay by inserting a hidden conservation law.

26.1 Witnessed composite role

The proton role is composite, but the theorem does not need a full theory of confinement. It needs only a target persistence role that is distinguishable from mesonic and leptonic final families by admitted boundary records and continuation histories. The witness family W_B supplies that distinction. The object B_{role} is therefore not a new primitive; it is an abbreviation for the conjunction of target witnesses.

26.2 Persistence under admissible scattering

The witness W_{scatter} is included to prevent overreading. A proton can scatter, dress, change momentum, bind into a nucleus, or interact electromagnetically while remaining the same target proton persistence class. Such changes are not proton decay because they preserve P_p under admissible histories. A decay claim must do more: it must remove the proton persistence role from the boundary-detectable final family.

26.3 Difference from mesonic and leptonic families

The witnesses W_{nonmes} and W_{nonlep} are needed because many proposed final states preserve electric charge and other obvious bookkeeping quantities. The target distinction is not charge

balance. It is loss of the baryonic composite persistence role. A final state such as a charged lepton plus mesons may preserve total electric charge but still lacks P_p . That absence is witnessed by the target boundary family, not by an unexplained baryon-number axiom.

26.4 No hidden selector

A hidden selector would say: this proton has an additional unwitnessed decay-enabled label. If the label changes no target witness, it is skin and cannot authorize a transition. If the label changes a witness, the target witness ledger has changed. If the label is retained as authoritative without witnesses, it is hidden load. Thus the witnessed B_{role} ledger blocks exactly the hidden-selector route available to a proposed counterexample.

26.5 Why ΔB_{role} is ledger motion

Because B_{role} is witness-defined, $\Delta B_{\text{role}} \neq 0$ means a change in the target witness package. A change in the target witness package is tensor motion. A transition whose tensor motion is not represented in $\mathcal{T}_{\text{tar}}$ is not a same-target morphism. A transition whose tensor motion is added to $\mathcal{T}_{\text{tar}}$ changes the occupant. This is the bridge from baryonic role loss to occupant-level stability.

27 Publication-readiness audit summary

The manuscript’s final theorem rests on a set of locally auditable bridges. This section records the status of each bridge and the failure route it blocks.

Bridge	Audit content	Failure route blocked
Proton persistence	Admits $[p]_{\text{persist}}$ as localized composite persistence role without deriving QCD numerics	“proton is assumed informally”
Baryonic role	Defines B_{role} through W_B witnesses	primitive baryon-number smuggling
Transition quotient	Requires same-target transitions to descend to quotient morphisms	representative-dependent transition rules
Ledger license	Requires quotient morphisms to be in $\mathcal{T}_{\text{tar}}$	bare mathematical functions as physical transitions
Event projection	Forces parent events to project to target boundary records	larger-domain escape hatch

EFT branch	Separates transport skin from primitive target operator	gauge-invariant operator objection
GUT branch	Classifies heavy mediators by induced target witness	GUT as same-target counterexample
Empirical interface	States what a positive event would change	non-falsifiability objection

Remark 27.1 (Publication standard). The proof exposes each bridge assumption directly. The table above is therefore part of the proof posture. A counterexample must identify which bridge fails and exhibit a standing-bearing object satisfying the counterexample burden. Without that object, the proposed escape reduces to same quotient, induced witness, or standing failure.

28 Nonclaims and positive claims

28.1 Nonclaims

The manuscript does not claim: no formal model can write a ΔB operator; no BSM theory exists; no UV completion can be studied; no quantum-gravity proposal can be meaningful; no experiment can report an anomalous event; AASC calculates proton lifetime; electric charge alone forbids proton decay; baryon number is a primitive metaphysical conservation law; all possible topological sectors are classified.

28.2 Positive claims

The manuscript does claim:

1. proton decay is a structural transition-role claim, not a numerical parameter claim;
2. the proton persistence class can be admitted as a target localized composite persistence role;
3. B_{role} is witness-defined, not primitive;
4. $\Delta B_{\text{role}} \neq 0$ implies transition-ledger motion unless already ledger-licensed;
5. same-target transitions must be ledger-licensed quotient morphisms;
6. no same-target proton-decay morphism is admitted in SM_{book} ;
7. larger-domain proton-decay claims either induce target witnesses, license no target decay, or fail standing;
8. occupant uniqueness blocks a non-equivalent target occupant containing a proton-decay role.

29 Conclusion

The theorem can be compressed into one audit sequent:

$$\begin{aligned}
& \Sigma_{\text{SM}} / \simeq_{\text{Adm}} \text{ structural target occupant} \\
& + \quad B_{\text{AASC}} / \simeq_{\text{Adm}} = \{[\text{SM}_{\text{book}}]\} \\
& + \quad [p]_{\text{persist}} \in \text{SM}_{\text{book}} \text{ as admitted proton persistence role} \\
& + \quad B_{\text{role}} \text{ witnessed by } W_B \\
& + \quad \Delta B_{\text{role}} \neq 0 \Rightarrow \Delta \mathcal{T}_{\text{tar}} \neq 0 \\
& + \quad \text{same-target transitions} \Rightarrow \text{ledger-licensed quotient morphisms} \\
& + \quad T_{p \rightarrow F}^{\Delta B} \notin \mathcal{T}_{\text{tar}}(\text{SM}_{\text{book}}) \\
& \Rightarrow \quad \neg \exists T_{p \rightarrow F}^{\text{Adm}} (\Delta B_{\text{role}} \neq 0).
\end{aligned}$$

For larger-domain proposals the controlling normal form is:

$$L^+|_{\text{tar}} = \begin{cases} \text{SM}_{\text{book}} & \Rightarrow \text{no target proton decay,} \\ \text{SM}_{\text{book}} + W_T^{\Delta B} & \Rightarrow \text{target ledger motion,} \\ \perp & \Rightarrow \text{standing failure or no target standing.} \end{cases}$$

Thus proton stability in this paper is neither an empirical accident nor merely the absence of a renormalizable term. It is the consequence of a structural fact: a baryon-violating proton-decay channel would be a new standing-bearing transition role, and the admissible target particle-physics occupant is already exhausted. A larger-domain theory may be proposed, but if it claims actual target proton decay it must induce the very transition witness that moves the target ledger. Same quotient means no target decay; induced witness means non-equivalent target occupant; witness-free assertion means standing failure. There is no admissible fourth route.

A Formal object ledger

Object	Meaning
$[p]_{\text{persist}}$	target proton persistence class
P_p	proton role package $(\Pi_{\text{loc}}, C_{\text{neutral}}, Q_{+1}, M_{\text{resp}}, \partial_{\text{detect}}, B_{\text{role}})$
W_B	baryonic persistence witness family
B_{role}	witnessed baryonic role coordinate
$\mathcal{T}_{\text{tar}}$	target transition ledger
$T_{\#}$	quotient-level transition morphism on $\mathcal{P}_{\text{tar}} / \sim_{\text{pers}}$

$\pi_{\text{tar}}(L^+, e)$	target projection of a parent-domain event
$O_{\Delta B}$	baryon-violating transition/operator class
$X_{\Delta B}$	heavy mediator or GUT-type carrier
$\text{QG}_{\Delta B}$	quantum-gravity or nonlocal parent carrier

B Expanded proton persistence certificate

The proton certificate has six components. First, Π_{loc} records localized persistence: the target class can be reused across boundary records as the same persistence object. Second, C_{neutral} records that the class is a color-neutral composite/hadronic target role rather than a raw colored representation slot. Third, Q_{+1} records the charge-lattice presentation, where charge magnitude is presentation of the charge lattice and not an independent coupling prediction. Fourth, M_{resp} records mass-response/inertial standing. Fifth, ∂_{detect} records boundary-detectable identity. Sixth, B_{role} records the witnessed baryonic composite persistence role.

A process that changes the skin of any component but preserves the witnesses does not destroy the proton class. A process that changes the boundary-detectable identity and removes the baryonic persistence witness package destroys the class and must be represented by a transition morphism. This is why scattering, dressing, binding-context change, and redescription are not proton decay, while ΔB_{role} loss is a transition-ledger question.

C Full quotient-morphism proof

Let $q : \mathcal{P}_{\text{tar}} \rightarrow \mathcal{P}_{\text{tar}} / \sim_{\text{pers}}$ be the quotient map. A raw transition rule $T : \mathcal{P}_{\text{tar}} \rightarrow \mathcal{P}_{\text{tar}}$ induces a quotient morphism iff there is a map $T_{\#}$ satisfying

$$T_{\#} \circ q = q \circ T.$$

This holds exactly when $x \sim_{\text{pers}} x' \Rightarrow T(x) \sim_{\text{pers}} T(x')$. If this condition fails, the output class depends on representative choice and the transition cannot be used as a same-target standing operation.

Suppose a proton-decay rule is claimed. If it does not satisfy the quotient condition, it is not an admissible same-target transition. If it satisfies the quotient condition but is not represented in $\mathcal{T}_{\text{tar}}$, then it is a quotient map without transition-ledger license. Using it changes the allowed morphism family. If it is represented in $\mathcal{T}_{\text{tar}}$, then it is already an admitted target transition. Since $\mathcal{T}_{\text{tar}}(\text{SM}_{\text{book}})$ has no ΔB_{role} proton-decay morphism, the admitted case is not proton decay. Hence every claimed same-target proton decay is unlicensed, ledger-expanding, or not decay.

D Full event-projection proof

Let e be a parent-domain event. The target projection $\pi_{\text{tar}}(L^+, e)$ is not a dynamical reduction assumption; it is the bookkeeping question whether the parent event can be represented as a target boundary record. There are four raw possibilities: no target record, proton-preserving target record, proton-destroying target record with witness, and proton-destroying target record without witness. The first is no target event. The second is no target proton decay. The third is induced target transition witness. The fourth is standing failure.

A proton-destroying target record with witness has the form

$$\partial_{\text{tar}}([p]_{\text{persist}}) \mapsto \partial_{\text{tar}}(F), \quad \partial_{\text{tar}}([p]_{\text{persist}}) \not\prec_{\text{Adm}} \partial_{\text{tar}}(F).$$

This is a target witness because boundary records are exactly the target interface by which persistence-class identity is compared. If the final record lacks P_p and B_{role} , the witness is a ΔB_{role} transition witness. Thus larger-domain physical decay cannot remain both target-decay-positive and target-ledger-inert.

E EFT operator branch exhaustion

For the operator $qqql/\Lambda^2$, the branch statuses are:

Reading	Required certificate	AASC status
Predictive EFT skin	parent domain, matching map, regime, error envelope, target projection	admissible as larger-domain transport representation; not same-target primitive
Primitive target term	same-target operator insertion	new operator ledger/load
Induced target witness	projected event loss of $[p]_{\text{persist}}$	target transition-ledger motion
Witness-inert notation	no target boundary record	no proton decay licensed

F Heavy-mediator and GUT branch exhaustion

For a mediator $X_{\Delta B}$, the target audit asks for $\text{Ind}_{\text{tar}}(X_{\Delta B})$. If the induced object is empty, no target decay is licensed. If the induced object is $O_{\Delta B}^{\text{tar}}$, the target transition ledger changes. If obtaining the induced object requires new representation, charge, anomaly, gauge, vacuum, or continuation structures, the parent theory is larger-domain load. If no matching or transport certificate is supplied, the mediator cannot carry target standing. These cases exhaust the target use of a GUT-type proton-decay carrier.

G Counterexample burden matrix

Claimed counterexample	Must supply	Failure point
Same-target $T_{p \rightarrow F}$	ledger-licensed quotient morphism with $\Delta B_{\text{role}} \neq 0$	no such morphism in SM_{book} ; adding it moves ledger
Gauge-invariant EFT term	transition standing beyond gauge syntax	primitive new load or transport image
GUT mediator	target projection and transition witness	induced witness implies target ledger motion
QG carrier	boundary-detectable target projection	no witness/no decay or ledger motion
Topological route	individual proton persistence loss	not target decay unless projection changes boundary record
Hidden decay label	invariant witness for label	no witness means selector failure
Empirical event	target persistence loss plus witness	target ledger motion or dependency failure

H Dependency matrix

Theorem component	Imported support	Function
Proton structural typing	Standalone SM Structure	operator and persistence ledgers are structural
Fixed-load no-completion	Structura	same-domain additions are skin or scope/load changes
Occupant uniqueness	Classification Theorem	final singleton target occupant
Transition morphism discipline	Classification + AASC kernel	standing quotient must be preserved under reuse
EFT primitive/transport distinction	Structura	higher-dimensional terms are transport images or new load

Charge-route exhaustion	Charge Rigidity and pattern for hidden CS6A selector/additive/extra-factor routing
Terminal normal form	Classification Theorem same quotient or induced witness classifier

I Scope locks and nonclaim ledger

This paper’s theorem is target-domain. It does not classify every mathematical theory; it classifies whether a proton-decay role is admissible in the unique target AASC particle-physics occupant. Larger-domain programs may be valuable. They are not same-target counterexamples unless they preserve the target quotient while producing a new standing-bearing proton-decay role. The quotient-morphism and event-projection theorems show that such a combination is impossible without ledger motion or standing failure.

J Event-projection details

The event-projection theorem prevents larger-domain language from becoming an escape hatch. This appendix expands the proof in a slower audit form.

J.1 Projection data

A parent-domain event claim is not itself a target-domain event. It must carry a projection package

$$\Pi_e = (L^+, e, \pi_{\text{tar}}, \partial_{\text{in}}, \partial_{\text{out}}, W_{\text{changed}}, \mathcal{T}_{\text{needed}}, \mathcal{D}_{\text{err}}, \mathcal{H}_{\text{hist}}),$$

where L^+ is the parent construction, e is the parent event record, π_{tar} is the map to the target comparison, ∂_{in} and ∂_{out} are target boundary-record extractions, W_{changed} records target witnesses changed by the event, $\mathcal{T}_{\text{needed}}$ records the transition-ledger entry needed to make the update target-admissible, $\mathcal{D}_{\text{err}}$ is any declared approximation/error discipline, and $\mathcal{H}_{\text{hist}}$ records how the event is used in subsequent target histories. A parent event without such a package can be meaningful in the parent theory without yet being a target proton-decay witness.

J.2 Four raw projection possibilities

Before fail-closed typing, there are four raw possibilities:

1. $\pi_{\text{tar}}(L^+, e)$ is undefined. The parent event has no target standing.
2. $\pi_{\text{tar}}(L^+, e)$ is defined, $W_{\text{changed}} = \emptyset$, and $\partial_{\text{out}} \simeq_{\text{Adm}} \partial_{\text{in}}$. The target proton class persists.

3. $\pi_{\text{tar}}(L^+, e)$ is defined, $W_{\text{changed}} \neq \emptyset$, and ∂_{out} is a non-proton final record. This is target witness induction.
4. $\pi_{\text{tar}}(L^+, e)$ is representative-dependent, selected after the fact, or asserts $\partial_{\text{out}} \not\vdash_{\text{Adm}} \partial_{\text{in}}$ while no changed target witness is supplied. This is standing failure or domain motion.

The event-projection theorem collapses these four raw cases to the three statuses used in the main text: no target decay, induced target transition witness, or standing failure/domain motion.

J.3 Why rare events still need target witnesses

A common objection says that a proton-decay event could remain invisible to the target transition ledger until it happens because the process is rare. AASC rejects that as a same-target claim. Rarity is a value-level or rate-level fact only after the transition channel has standing. If the channel is not in the target transition ledger, then assigning a tiny nonzero rate already presupposes the missing transition carrier. The order is:

$$\text{transition role admitted} \quad \Rightarrow \quad \text{rate or lifetime can be assigned.}$$

The reverse order is inadmissible:

$$\text{choose tiny rate} \quad \nRightarrow \quad \text{transition role admitted.}$$

Thus a rare larger-domain event must still project to a target witness if it is to count as target proton decay.

J.4 Boundary-record change as the minimal witness

The theorem does not require a full dynamical explanation of the event at target level. It only requires a minimal witness: the boundary record before and after the event must distinguish proton persistence from non-proton final persistence. Formally,

$$\partial_{\text{tar}}([p]_{\text{persist}}) \not\vdash_{\text{Adm}} \partial_{\text{tar}}(F).$$

This inequality is already enough to induce a target transition witness when paired with $W_{\text{changed}} \neq \emptyset$. If the parent theory supplies this inequality, it has moved the target transition ledger. If it does not, the parent theory has not licensed target proton decay.

J.5 Projection and occupant uniqueness

The event-projection theorem does not itself prove occupant uniqueness. It ensures that larger-domain claims enter the target proof at the correct point. Once a parent event induces a target transition witness, the target occupant is no longer merely SM_{book} ; it is

$SM_{\text{book}} + W_T^{\Delta B}$. Occupant uniqueness then excludes this as a second target occupant. If the parent event induces no such witness, occupant uniqueness is not needed; there is simply no target proton-decay transition.

Theorem J.1 (Expanded event-projection trichotomy). *Let Π_e be a projection package for a parent event. Exactly one of the following holds after quotienting skin and representative choices:*

- (i) $\partial_{\text{tar}}([p]_{\text{persist}})$ is preserved and $W_{\text{changed}} = \emptyset$: no target proton decay;
- (ii) $\partial_{\text{tar}}([p]_{\text{persist}})$ is replaced by a non-equivalent final boundary record and $W_{\text{changed}} \neq \emptyset$: induced target transition witness;
- (iii) no well-defined target boundary record or changed-witness package exists: standing failure or nonprojection.

Proof. The target boundary record is an equivalence class under admissible target redescription. A parent projection either maps the input class to itself without changed target witnesses, maps it to a distinct target class with changed target witnesses, or fails to produce a well-defined target class and changed-witness package. These are mutually exclusive and exhaustive for a quotient-valued projection. The first case preserves proton persistence; the second supplies a witness of proton-persistence loss; the third supplies no target standing. Representative-dependent maps are included in the third case because they do not descend to the quotient. $\square$

K Parent-to-target transport audit

This appendix records the generic transport certificate required whenever a larger-domain construction is used to support a target proton-decay claim.

K.1 Transport certificate schema

A parent-to-target transport certificate for proton decay is

$$\text{Cert}_{L^+ \rightarrow \text{tar}} = (C_1, C_2, C_3, C_4, C_5, C_6, C_7),$$

where:

1. C_1 : parent carrier is specified, not merely named;
2. C_2 : target regime and retained degrees are declared;
3. C_3 : matching or projection map is explicit;
4. C_4 : error envelope or approximation status is carried;
5. C_5 : target boundary-record extraction is defined;

6. C_6 : target transition witness status is classified;
7. C_7 : same quotient, induced witness, or standing failure is reported.

K.2 Why transport completeness matters

If the parent loses data relevant to target standing, the target event cannot be reused as a stable proton-decay claim. If the target projection invents data not present in the parent or target ledgers, it is a selector. If the target projection changes the standing quotient, it is target ledger motion. Transport completeness is therefore not decorative; it is the condition under which a parent-domain event can be compared with a target-domain persistence class.

Proposition K.1 (Transport certificate necessity). *A larger-domain proton-decay claim without $\text{Cert}_{L^+ \rightarrow \text{tar}}$ is not a same-target counterexample.*

Proof. A same-target counterexample must show that the larger-domain carrier acts on $[p]_{\text{persist}}$ while preserving the target standing quotient and transition ledger. Without the certificate, no such action has been exhibited. The claim is either parent-domain only, notation, or a request to change the target domain. $\square$

K.3 Same quotient and induced witness

The same-quotient case is often misunderstood. It does not mean the parent theory is false or trivial. It means that, when restricted to the target comparison, the parent has not supplied a target proton-decay witness:

$$L^+|_{\text{tar}} = \text{SM}_{\text{book}}.$$

The induced-witness case means:

$$L^+|_{\text{tar}} = \text{SM}_{\text{book}} + W_T^{\Delta B}.$$

This is exactly the object excluded by occupant uniqueness as a same-target occupant. The classification is therefore not circular: the parent restriction is computed first, and occupant uniqueness is applied only after the target restriction is known.

K.4 Failure modes

Failure mode	Meaning
No parent carrier	The larger-domain source is named but not structurally specified.
No matching map	The route from parent dynamics to target bookkeeping is absent.

No boundary record	The event cannot be compared with $\partial_{\text{tar}}([p]_{\text{persist}})$.
Representative-dependent projection	Target result depends on skin or path choice.
No error envelope	EFT or approximation status is not reportable.
New witness untyped	The parent induces target data but does not declare target ledger motion.
Post-selection	Only successful decay histories are retained as admissible.

L EFT transport certificate for baryon-violating operators

A dimension-six operator such as $qqql/\Lambda^2$ can appear in at least three scientific contexts. The AASC audit separates them.

L.1 Context A: formal operator basis

A formal operator basis lists all expressions compatible with a chosen syntax and symmetry screen. This is valuable for EFT bookkeeping. It does not by itself assert that every listed expression has target transition standing. In this context $qqql/\Lambda^2$ is a basis element, not a proton-decay role.

L.2 Context B: matched EFT from a parent

A matched EFT inherits standing from a parent carrier and a matching calculation. The admissible object is not merely $qqql/\Lambda^2$, but

$$(L^+, \text{match}, \mathcal{D}_{\text{err}}, \mathcal{O}_{\Delta B}^{(6)}, \pi_{\text{tar}}).$$

This object can be meaningful, but its target proton-decay claim still passes through event projection. If π_{tar} gives a ΔB boundary-record change, the target transition ledger has moved.

L.3 Context C: primitive low-energy addition

If the operator is simply appended to the target occupant as a primitive, the target operator ledger changes immediately:

$$\mathcal{O}_{\text{SM}} \mapsto \mathcal{O}_{\text{SM}} \cup \{\mathcal{O}_{\Delta B}^{(6)}\}.$$

That is not same-occupant redescription. It is $+\Delta B$ load.

Theorem L.1 (EFT certificate theorem). *A baryon-violating EFT operator has AASC transition standing only as part of a certificate package containing source, matching, error*

envelope, target projection, and transition-witness classification. Without this package it is formal syntax or primitive new load.

Proof. The operator basis alone does not define a target persistence-class morphism. A matched EFT defines such a morphism only after target projection. A primitive addition changes the operator ledger. Therefore the certificate package is necessary for transition standing. $\square$

L.4 Why the coefficient does not help

A coefficient such as $1/\Lambda^2$ is a rate-scale or suppression datum only after the operator has standing. It cannot bootstrap the operator into standing. A zero coefficient is absence of the channel at the level of that presentation. A nonzero coefficient presupposes the channel. Thus the coefficient cannot be the AASC source of proton decay.

M Full route exhaustion table

Route	Candidate object	Target projection	Blocking theorem
Gauge redescription	gauge representative change	preserves $[p]_{\text{persist}}$	Redescription exhaustion
Basis/flavor change	coordinate relabeling	skin	Redescription exhaustion
Charge normalization/sign	charge presentation	no transition	Charge-lattice rigidity
Finite correction	coefficient/rate	channel absent	Decay channel versus rate
Yukawa/mass insertion	mass-response value	no $O_{\Delta B}$	Structural typing
Primitive $qqql/\Lambda^2$	new operator	new ledger	EFT Branch Exhaustion
Transport-image $qqql/\Lambda^2$	matched EFT term	induced-target trichotomy	Event projection
GUT mediator	$X_{\Delta B}$	witness or none	Heavy-Mediator Branch Exhaustion
Extra $U(1)$	extra charge direction	larger charge scope	Charge rigidity

Extra scalar multiplet	altered Yukawa/neutralty skeleton		larger scope	Charge rigidity / Structura
Extra chiral matter	altered ledger	anomaly	larger scope	Structura
QG boundary carrier	QG _{ΔB}		witness or none	Event projection
Nonlocal histories carrier	parent-domain history	his-	witness or none	Induced Target Transition
Sphaleron / topological	topological process		target loss or not	Topological Route Boundary
Hidden decay label	unwitnessed tag		selector failure	Baryonic Role Standing
Post-selection repair	generate/filter route		repair	AASC no-repair discipline
Empirical event	boundary record		ledger motion or failure	Empirical Interface

N Publication audit notes

This appendix records the intended referee-audit reading.

N.1 What the paper proves

It proves that no proton-decay transition role is admissible in the unique target AASC particle-physics occupant. It also proves that a larger-domain proton-decay claim must induce target transition standing or fail to license target proton decay.

N.2 What the paper deliberately does not prove

It does not compute hadron dynamics, proton lifetime, confinement, or a full QCD spectrum. It does not classify every mathematical theory. It does not deny that larger-domain models may be formulated. It does not assert primitive baryon conservation.

N.3 Most likely objections and exact answers

Objection	Exact answer
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“The proof is tautological.”	The quotient-morphism theorem is not definitional; it proves that same-target transitions must descend to ledger-licensed maps on standing classes.
“A GUT event can be rare and still real.”	Rarity is a rate property after transition standing. A rare event still needs target projection to count as target proton decay.
“EFT operators are legitimate.”	Yes, as formal bases or matched EFT skins. They are not same-occupant primitives; if matched, they trigger event projection.
“You smuggle in baryon conservation.”	B_{role} is witness-defined by W_B , and the theorem uses role loss as ledger motion, not as a primitive conservation law.
“Quantum gravity may violate global symmetries.”	Then it must project to a target boundary-record change. That is induced target witness motion, not same-target redescription.
“An observation would refute you.”	A certified same-target transition preserving all ledgers would refute the theorem or its dependencies. A transition requiring new witnesses is larger-domain ledger motion.

N.4 Final audit sequent

$$\begin{aligned}
& [p]_{\text{persist}} \in \text{SM}_{\text{book}}, \quad B_{\text{role}} \text{ witnessed by } W_B, \quad \Delta B_{\text{role}} \neq 0 \Rightarrow \Delta \mathcal{T}_{\text{tar}} \neq 0, \\
& \text{same-target transitions} \Rightarrow \text{ledger-licensed quotient morphisms}, \quad T_{p \rightarrow F}^{\Delta B} \notin \mathcal{T}_{\text{SM}}, \\
& B_{\text{AASC}} / \simeq_{\text{Adm}} = \{[\text{SM}_{\text{book}}]\} \Rightarrow \neg \exists T_{p \rightarrow F}^{\text{Adm}} (\Delta B_{\text{role}} \neq 0).
\end{aligned}$$

References

- [1] Amos Jay Maley. *The Standalone Standard Model Structure from First Principles: An AASC Derivation of Gauge-Product, Representation, Higgs/Bridge, Yukawa, Charge, Anomaly, Vacuum, and Parameter-Status Structure*. Uploaded manuscript, May 20, 2026.
- [2] Amos Jay Maley. *Structura Ex Necessitate Standardis Modelis: A relative closure theorem for the Standard-Model bookkeeping interior under fixed load, fixed domain, and certificate-maximal same-domain transport*. Uploaded manuscript, April 29, 2026.
- [3] Amos Jay Maley. *A Classification Theorem for Admissible Standard-Model Structures: Uniqueness Under Standing-Preserving and Transport Constraints*. Uploaded manuscript, May 5, 2026.

- [4] Amos Jay Maley. *Standard-Model Realization Rigidity in the Admissible Gauge Arena*. Uploaded manuscript, May 2026.
- [5] Amos Jay Maley. *Rigidity of the Standard-Model Charge Lattice: A Fixed-Scope Reconstruction Theorem for Charge Quantization*. Uploaded manuscript, May 2026.
- [6] Amos Jay Maley. *Fixed-Scope Standard-Model Charge-Lattice Singleton Extraction*. Uploaded manuscript, May 2026.
- [7] Amos Jay Maley. *Pilot Metric-Relative Standard Model Reconstruction*. Uploaded manuscript, May 2026.
- [8] Amos Jay Maley. *The Standard Model as the Unique Minimal Standing-Preserving Bookkeeping Structure*. Uploaded manuscript, 2026.
- [9] Howard Georgi and S. L. Glashow. “Unity of All Elementary-Particle Forces.” *Physical Review Letters* **32**, 438–441 (1974). doi:10.1103/PhysRevLett.32.438.
- [10] Steven Weinberg. “Baryon- and Lepton-Nonconserving Processes.” *Physical Review Letters* **43**, 1566–1570 (1979). doi:10.1103/PhysRevLett.43.1566.
- [11] Frank Wilczek and A. Zee. “Operator Analysis of Nucleon Decay.” *Physical Review Letters* **43**, 1571–1573 (1979). doi:10.1103/PhysRevLett.43.1571.
- [12] Steven Weinberg. “Varieties of Baryon and Lepton Nonconservation.” *Physical Review D* **22**, 1694–1700 (1980). doi:10.1103/PhysRevD.22.1694.
- [13] Norisuke Sakai and Tsutomu Yanagida. “Proton Decay in a Class of Supersymmetric Grand Unified Models.” *Nuclear Physics B* **197**, 533–542 (1982). doi:10.1016/0550-3213(82)90457-6.
- [14] Pran Nath and Pavel Fileviez Perez. “Proton Stability in Grand Unified Theories, in Strings, and in Branes.” *Physics Reports* **441**, 191–317 (2007). doi:10.1016/j.physrep.2007.02.010.
- [15] Andrew Kobach. “Baryon Number, Lepton Number, and Operator Dimension in the Standard Model.” *Physics Letters B* **758**, 455–457 (2016). doi:10.1016/j.physletb.2016.05.050.
- [16] P. S. B. Dev et al. “Searches for Baryon Number Violation in Neutrino Experiments: A White Paper.” *Journal of Physics G: Nuclear and Particle Physics* **51**, 033001 (2024). doi:10.1088/1361-6471/ad1658.
- [17] Volodymyr Takhistov. “Experimental Tests of Baryon and Lepton Number Conservation.” arXiv:2602.09097 (2026).

- [18] Super-Kamiokande Collaboration. “Search for Proton Decay via $p \rightarrow e^+\pi^0$ and $p \rightarrow \mu^+\pi^0$ with an Enlarged Fiducial Volume in Super-Kamiokande I–IV.” *Physical Review D* **102**, 112011 (2020). arXiv:2010.16098. doi:10.1103/PhysRevD.102.112011.
- [19] Super-Kamiokande Collaboration. “Search for Proton Decay via $p \rightarrow \nu K^+$ Using 260 kiloton year Data of Super-Kamiokande.” *Physical Review D* **90**, 072005 (2014). arXiv:1408.1195. doi:10.1103/PhysRevD.90.072005.
- [20] F. Takahashi et al. (Particle Data Group). “Review of Particle Physics.” *International Journal of Modern Physics A* **41**, 2630011 (2026); online listings at PDG/pdgLive.